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Dynamic Multi-product Procurement With Joint and Individual Setup Costs: Theory and Insights

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Abstract

In practice, it is common, especially for online retailers, to bundle different products together during procurement to save transportation and handling costs. It is important to understand and theorize how to manage dynamic procurement by taking advantage of joint ordering in the presence of joint and individual setup costs. In this article, we characterize the structure of optimal policy for a periodic review multiproduct inventory system with multiple setup costs, including a joint setup cost and an individual setup cost for each product. By proposing the notion of $(\mathcal{K}, \boldsymbol{\eta})$-quasi-convexity, we show that an optimal procurement policy for such a system follows the so-called $(\sigma, \omega, \boldsymbol{S})$ policy when demands increase stochastically over time: order up to $\boldsymbol{S}$ for states in the region σ, do not order for states in the region ω, and order certain quantities for states in neither σ nor ω. To better understand the optimal policy, we provide the bounds for the optimal order-up-to levels and the boundary sets of the optimal policy. Under the convex single-period inventory costs, we also provide a lower bound deterministic system which can be asymptotically optimal as the coefficient of variations decreases to zero. Leveraging these operational insights, we propose five simple heuristic policies: the independent (s, S) policy, vector $(\boldsymbol{s}, \boldsymbol{S})$ policy, linear interpolation $(\boldsymbol{s}, \boldsymbol{S})$ policy, the deterministic approximation, and the weighted deterministic approximation policy. Extensive numerical experiments indicate that the last three heuristics perform well. In particular, the weighted deterministic approximation policy, whose average performance gap is $< 1\%$, dominates the others in almost all our numerical experiments. Finally, we show that how our results can be extended to systems with more complex setup cost functions, such as time-varying, set-based, and quantity-dependent setup costs.

Keywords

Inventory/production systems, $(\mathcal{K}, \boldsymbol{\eta})$-quasi-convexity, multiple setup costs, multiple products

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1 Introduction

Inventory management can improve service rates and hedge demand uncertainty, which has become an important component of the modern economy. It is estimated that the total investment in inventory for a quarter in the United States could reach \$1.57 trillion (Nahmias and Cheng, 2009). Effective inventory and procurement strategy is crucial for many firms to succeed in volatile business environments and against global competition. In practice, procurement often involves more than one product, and replenishment is often needed for multiple products at the same time (Feng et al., 2015). For instance, online retailers like Amazon.com and JD.com commonly need to manage hundreds of categories of products simultaneously. In such cases, there is typically a joint setup cost for the bundle of products to be ordered, as well as an individual setup cost for each product. It is therefore critical for firms to understand the procurement strategies and obtain operational insights in the presence of joint and individual setup costs.

There is an extensive literature focusing on determining optimal inventory policies to effectively manage complex

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ordering costs. However, most of the existing literature primarily addresses single-product systems and different ordering cost scenarios, while the optimal policies for systems with multiple products and multiple setup costs are still an open area of study (Perera and Sethi, 2022). Nevertheless, systems involving multiple products and multiple setup costs are quite common in joint ordering situations, such as those encountered by online sellers, wholesalers, Amazon.com, and JD.com. In joint ordering, there is a joint setup cost associated with ordering a bundle of products, which includes fixed transportation and handling expenses. Typically, when placing a new order, there is a separate handling and hassle cost associated with each individual product. In certain cases, supply contracts are structured as two-part tariff contracts, where each order incurs a fixed procurement cost along with a variable cost.

For multi-product systems, there may be a product-independent joint fixed cost incurred for dispatching an empty truck. Additionally, there could be a fixed cost for each product due to extra loading requirements, special containers, or specific supply contracts. Consider the example of a retailer with a warehouse that replenishes from multiple suppliers. In this scenario, the transportation cost from each supplier can be considered as an individual setup cost, while the transportation cost from the warehouse to the retailer can be interpreted as the joint setup cost (Feng et al., 2015).

With the joint setup costs, it is conceivable that ordering interactions exist between different products. We argue that the operational insights obtained under single-product settings typically cannot shed light on how to manage multi-product supply chains due to the presence of the joint and individual setup costs. In this article, we investigate the optimal procurement policy for a periodic-review multi-product inventory system with a joint setup cost and individual setup costs under a finite horizon. We assume demands for different products are independent in different periods, but the demands can be correlated in the same period, for example, demands for different products can be complementary or result from customer choice (e.g., horizontally differentiated products). Unmet demands are backlogged. The procurement decisions for all products are made simultaneously at the beginning of each period. The ordering costs include a variable cost for each product, a joint setup cost, and an individual setup cost for each product. The joint setup cost is incurred if a positive order is placed for any product, and an individual setup cost is incurred if any positive order is placed for the corresponding product. The objective is to determine the optimal procurement strategy to minimize the total expected discounted cost throughout the planning horizon.

However, it is technically challenging to analyze the multi-product inventory problem with multiple setup costs. To tackle the issue, we propose a new class of functions, the $(\mathcal{K}, \boldsymbol{\eta})$-quasi-convex functions. Under the condition that the global minima of the single-period costs are increasing over time, we show that the value function in each period is $(\mathcal{K}, \boldsymbol{\eta})$-quasi-convex. By exploiting the proposed $(\mathcal{K}, \boldsymbol{\eta})$-quasi-convex functions, we show that the optimal policy for the above inventory system can be described by the so-called $(\sigma, \omega, \boldsymbol{S})$ inventory policy, where σ and ω are two sets and $\boldsymbol{S}$ is a vector. Under the optimal policy, orders are not placed for the inventory states in ω, and orders up to $\boldsymbol{S}$ are placed for the inventory states in σ. Finally, orders up to certain quantities (but not necessarily to $\boldsymbol{S}$) are placed for the inventory states in neither ω nor σ.

We obtain the ordering monotonicity properties that further characterize the structure of the reorder set based on partial orders: if it is optimal to order under a given inventory state, then it must be optimal to order up to the same level under any other inventory state which is less than the former under a class of partial orders. For example, given an inventory state in σ, any inventory state vector that is component-wise less than that state vector with respect to a class of partial orders must be in σ as well, that is, under which it is optimal to order up to $\boldsymbol{S}$. The monotonicity property not only applies to the set σ, but also applies to those states that do not order up to $\boldsymbol{S}$, that is, the set $\theta \setminus \sigma$, where θ is the complementary set of ω. To better understand the optimal policy, we then derive the explicit bounds for the parameters of the optimal policy, including the order-up-to levels and boundaries of sets σ and $\theta \setminus \sigma$. Furthermore, under the convex single-period inventory costs, we derive a lower bound deterministic system which can be asymptotically optimal as the coefficient of variation of demands decreases to zero. The lower bound system is indeed a multi-product lot-sizing problem which is well-studied in the literature. Note that the system under any feasible heuristic is an upper bound system. Thus, the difference between the system cost under a feasible heuristic and the lower bound system is the maximum performance gap for the heuristic. These bounds and monotonicity properties are useful in computing the optimal policy and designing effective heuristics.

To the best of our knowledge, our paper is the first to analytically characterize the structure of optimal policies for inventory systems with multiple products and multiple setup costs. The notion of the $(\mathcal{K}, \boldsymbol{\eta})$-quasi-convexity generalizes the results by Veinott (1966) and Kalin (1980) to multiple products and any subadditive setup costs. Kalin (1980) only considers the joint setup cost K, which is a constant that is independent of the products. By contrast, we allow the setup cost to be a function of the order quantities and dependent on each product. Because of the presence of individual setup costs, the partial order by Kalin (1980) that can only characterize the no-order set is not applicable in our setting. We then propose a novel partial order that induces an ordering monotonicity property, which can directly characterize the ordering behavior for states that should be ordered. The monotonicity property however is absent by Kalin (1980). As a result, our optimal policy is different from by Kalin (1980). In fact, the latter is a special case of our optimal policy. Furthermore, our partial order allows us to derive the explicit bounds for the optimal parameters of the optimal policy, which are also new in the literature.

By leveraging these structural properties and bounds of the optimal $(\sigma, \omega, \boldsymbol{S})$ policy, we propose five simple heuristics: the independent $(\boldsymbol{s}, \boldsymbol{S})$ policy, the vector $(\boldsymbol{s}, \boldsymbol{S})$ policy, the linear interpolation $(\boldsymbol{s}, \boldsymbol{S})$ policy, the deterministic approximation policy, and the weighted deterministic approximation policy. The independent $(\boldsymbol{s}, \boldsymbol{S})$ policy orders each product following an item-wise (s, S) policy, and the vector $(\boldsymbol{s}, \boldsymbol{S})$ policy orders all products simultaneously. The linear interpolation $(\boldsymbol{s}, \boldsymbol{S})$ policy approximates the frontier of the joint order region σ by lines or hyperplanes. By contrast, the deterministic approximation policy first solves an lower bound system and utilizes the corresponding ordering sets as a surrogate of the optimal $(\sigma, \omega, \boldsymbol{S})$ policy. The weighted deterministic approximation policy further improves the deterministic approximation policy by constructing a state-dependent order-up-to levels using a linear combination of the order-up-to levels of the lower bound system and an upper bound of $\boldsymbol{S}$. We also design greedy and exhausted search algorithms to specify the parameters of the heuristics, for example, the reorder points and order-up-to levels.

Our extensive numerical experiments indicate that the last three heuristics, that is, the linear interpolation $(\boldsymbol{s}, \boldsymbol{S})$ policy, the deterministic approximation policy, and the weighted deterministic approximation policy, perform extremely well and the weighted deterministic approximation policy dominates others in almost all numerical experiments. The average performance gap of the linear interpolation $(\boldsymbol{s}, \boldsymbol{S})$ policy is $< 3.58\%$ whereas the average performance gaps of the deterministic approximation policy and the weighted deterministic approximation policy are both $< 1\%$ in all experiments. Hence, we recommend these three heuristics to manage complex inventory systems with multiple setup costs in practice because of their simplicity and efficiency. Specifically, if the problem's deterministic counterpart is not very complex to solve, then the (weighted) deterministic approximation policy is strongly recommended; otherwise, the linear interpolation $(\boldsymbol{s}, \boldsymbol{S})$ policy may be the best choice.

We also discuss how to extend our results to more general settings, such as (i) cases when the inventory costs are time-varying, and (ii) cases when there are additional joint setup costs for a subset of products. In addition, the $(\boldsymbol{\mathcal{K}}, \boldsymbol{\eta})$-quasi-convexity can be extended to cases with additional batch-size related setup costs for individual products.

The rest of the article is organized as follows. Section 2 reviews the related literature. Section 3 presents the formulation of our problem. Section 4 introduces the related definitions and properties on $(\boldsymbol{\mathcal{K}}, \boldsymbol{\eta})$-quasi-convex functions. Section 5 characterizes the structure of the optimal procurement policy for our system by leveraging the $(\boldsymbol{\mathcal{K}}, \boldsymbol{\eta})$-quasi-convexity. Section 6 provides bounds for the optimal parameters of the optimal policy and a lower bound system that is asymptotically optimal. Section 7 proposes five simple heuristics and investigates their performance via extensive numerical experiments. Section 8 describes how we can extend our results and Section 9 concludes. All proofs, additional analyses, and detailed numerical results can be found in the Appendices in the E-Companion.

2 Related Literature

We contribute to the literature on inventory control with fixed ordering costs or setup costs.

The earlier works address optimal policies for single product inventory systems with a single fixed ordering cost. One of the classical results is that an (s, S) policy is optimal for such a system, that is, order up to S if the inventory level falls below s, and otherwise order nothing. Scarf (1960) is the first paper to show the optimality of the (s, S) policy by leveraging the so-called K-convex functions. Veinott (1966) provides alternative conditions and new proofs for the optimality of the (s, S) policy by assuming one-period expected holding/backordering cost to be quasi-convex. It is well-known that his conditions do not imply and are not implied by conditions by Scarf (1960). Recently, Chen et al. (2015) showed when the (s, S) policy is still optimal under time evolving demands and cost parameters.

The literature is extensive on optimal policies for inventory systems with different types of fixed costs in the single-product setting. Caliskan-Demirag et al. (2012) investigated a periodic-review single-product system with two quantity-dependent costs K_1, K_2 and partially characterize the optimal policy under conditions (1) $K_1 \leq 2K_2$ and (2) $K_1 \leq K_2$, by introducing the notion of C-(K_1, K_2)-convexity and strong-K-convexity, respectively. These notions are generalizations of the K-convexity by Scarf (1960) and similar notions such as CK-convexity and (C, K)-convexity can be found in studies on single product capacitated inventory systems; see for example, Gallego and Scheller-Wolf (2000) and Shaoxiang (2004). Lu and Song (2014) introduced the strong-(K, c, q)-convexity to consider an inventory system with a piecewise linear convex cost. Lu et al. (2016) proposed the K-approximate convexity to investigate the inventory system with incomplete demand information and Lu et al. (2018) further applied this notion to develop heuristic policies for the inventory system with a general piecewise linear cost. Li and Yu (2012) showed the quasi-convexity for lost sales inventory systems with fixed costs. Benjaafar et al. (2018) analyzed inventory systems with concave ordering costs by proposing c-convexity. Federgruen et al. (2020) proposed the notion of strong-(C_1K_1, C_2K_2)-convexity which synthesizes some common K-convexity concepts in the literature. Unfortunately, since K-convexity in $\mathbb{R}^n$ (similarly its variants, for example, C-(K_1, K_2)-convexity) usually cannot be preserved under the minimization operator (Gallego and Sethi, 2005), these notions may not be readily extended to deal with multiple products. We also refer readers to Perera and Sethi (2022) for the recent survey on the (s, S)-type policy.

A stream of closely related literature focuses on managing multiple product inventory systems with a single joint setup cost. Johnson (1967) considers an infinite horizon multiple product system with a constant joint fixed ordering cost K. He shows that the $(\sigma, \boldsymbol{S})$ policy is optimal for the above system, that is, ordering up to $\boldsymbol{S}$ for states in σ, using the policy improvement method in Markov decision processes. Similarly, Ohno and Ishigaki (2001) showed that a $(\sigma, \boldsymbol{S})$ policy is optimal for a continuous review inventory problem with Poisson demands and investigate the performance of three heuristic policies. Kalin (1980) also shows that a $(\sigma, \boldsymbol{S})$ policy is optimal for the same problem by Johnson (1967) by introducing the notion of $(K, \boldsymbol{\eta})$-quasi-convexity that extends the quasi-convexity of Veinott (1966). The set $(\sigma, \boldsymbol{S})$ policy by Kalin (1980) indicates that one should order up to $\boldsymbol{S}$ for inventory states $\boldsymbol{x} \leq \boldsymbol{S}, \boldsymbol{x} \notin \sigma$. Liu and Esogbue (2012) generalized the K-convexity by Scarf (1960) to $\mathbb{R}^n$ and partially show the optimality of an $(\boldsymbol{s}_t, \boldsymbol{S}_t)$ policy in a finite horizon setting, under the conditions that the initial inventory level $\boldsymbol{x}_1 \leq \boldsymbol{S}_1$ and that $\boldsymbol{S}_t$ increases in time t. By contrast, we consider multiple product inventory systems, with multiple setup costs, including a joint setup cost incurred whenever any product is ordered and an individual setup cost for each product whenever this type of product is ordered. As Perera and Sethi (2022) point out, the optimal policy for such complex inventory system still remains open in the literature. We then provide the first theoretical result on the optimal policy for such a system.

From a technical perspective, our work is partially inspired by Veinott (1966) and Kalin (1980). Specifically, our work generalizes Kalin (1980) by considering individual setup costs for each product and generalizing $(K, \boldsymbol{\eta})$-quasi-convexity, where K by Kalin (1980) is a constant, while $\boldsymbol{\mathcal{K}}$ in ours is a generalized indicator function of order quantities. Due to the presence of individual setup costs, the partial order by Kalin (1980) that only characterizes the no-order set is not applicable in our setting. We thus proposed a new partial order to prove our optimal policy. The ordering monotonicity properties that we proved can directly characterize the ordering behavior (set), which is absent by Kalin (1980). As a result, the optimal policy by Kalin (1980) is a special case of our optimal policy. Finally, the $(\boldsymbol{\mathcal{K}}, \boldsymbol{\eta})$-quasi-convexity holds for subadditive setup costs.

Another stream of literature develops simple heuristics and efficient algorithms to manage complex inventory systems with multiple setup costs. For example, Balintfy (1964) proposes the "can-order" policy or (s, c, S) policy for a continuous review system with deterministic demands and the objective to minimize the total cost. Federgruen et al. (1984) offered algorithms to compute (s, c, S) policy for continuous review inventory systems with compound Poisson demands to minimize the long-run average cost per unit time. However, such a policy is usually not optimal even in a simple setting with only two products (Ignall, 1969). With the same objective as by Federgruen et al. (1984), Atkins and Iyogun (1988) instead suggested a periodic replenishment policy for a continuous review multiple product system with multiple setup costs. Shang (2008) proposes a simple two-step heuristic to find the base order quantities for serial inventory systems with echelon-wise fixed order costs. Recently, Feng et al. (2015) proposed a heuristic called (s, c, d, S) policy for a continuous review system and correlated demands to minimize the total discounted expected cost. Other heuristic policies in the literature include the $P(s, S)$ policy, $Q(s, S)$ policy (e.g., Viswanathan, 1997), and (Q, S, T) policy (e.g., Özkaya et al., 2006), among others. In contrast to previous work, we aim to analytically show the optimality of the $(\sigma, \omega, \boldsymbol{S})$ policy for periodic-review multiple-product inventory systems with multiple setup costs. We also propose five simple heuristics inspired by the structure and bounds of the optimal policy. Extensive numerical experiments indicate that three heuristics performs extremely well whose average performance gap can be < 1%.

3 Model and Formulation

We consider a periodic-review inventory system with n different products over a finite planning horizon with T periods. Products and periods are indexed by $i = 1, \ldots, n$ and $t = 1, \ldots, T$, respectively. The demand $D_{i,t}$ for each product i in each period t is random with a continuous marginal distribution function $F_{i,t}$ and a finite mean $d_{i,t} = \mathbb{E}[D_{i,t}]$. Demands are independent across periods. We also allow demands $D_{i,t}, i = 1, \ldots, n$ for different products in each period to be correlated with a joint distribution function F_t and there is no restriction on the types of demand correlation.

At the beginning of each period t, non-negative order quantities $(z_{1,t}, \ldots, z_{n,t})$ of the n products are placed simultaneously to fulfill the stochastic demands $(D_{1,t}, \ldots, D_{n,t})$. We assume that inventory is replenished from an outside supplier immediately with ample stock, that is, there are no capacity constraints. We assume zero lead times and the case with fixed lead times can be treated using a standard state transformation. Then demands for the n products are realized and fulfilled by the on-hand inventory, if any is available; otherwise all unmet demands are backlogged. Leftover inventory is carried over to the next period.

In each period t, let $x_{i,t}$ be the net inventory level (or simply the inventory level), that is, the amount of inventory minus the amount of backorders, for product $i, i = 1, \ldots, n$. Following the literature convention, let $\boldsymbol{x}_t = (x_{1,t}, \ldots, x_{n,t})$ be the vector of the net inventory levels for all products at the beginning of period t and $\boldsymbol{y}_t = (y_{1,t}, \ldots, y_{n,t})$ be the vector of the order-up-to levels, that is, the post-ordering inventory levels. Similarly, let $\boldsymbol{z}_t = (z_{1,t}, \ldots, z_{n,t}), \boldsymbol{D}_t = (D_{1,t}, \ldots, D_{n,t}), \boldsymbol{d}_t = (d_{1,t}, \ldots, d_{n,t})$ be the order quantities, demands and mean demands for all products in period t, respectively. To simplify our notation, we suppress subscript t in our variables whenever no ambiguity arises.

As mentioned earlier, in procurement, it is not uncommon to reduce costs by bundling different products for transportation and handling. Moreover, typically, there is a handling and hassle cost for each individual product if we place a new order. In many cases, the supply contracts are determined by two-part tariff contracts (Shin and Tunca, 2010), that is, for each order there is a fixed procurement cost and a variable cost. To capture these rich settings of procurement costs, we consider two types of costs in each period, including setup costs and variable ordering costs. The setup costs consist of a joint setup cost for all products and an individual setup cost for each product. The variable cost for each product is due to its prevailing price and the amortized linear transportation cost. The joint setup cost is incurred if any order for any product is placed, while an individual setup cost and a variable cost for a product are incurred only if an order is placed for the product. Define

$$\delta(z) := \begin{cases} 1 & \text{if } z > 0, \\ 0 & \text{otherwise,} \end{cases}$$

as the indicator function, and

$$\mathcal{K}(\boldsymbol{z}) = K_0 \delta\left(\sum_{i=1}^{n} z_i\right) + \sum_{i=1}^{n} K_i \delta(z_i), \tag{1}$$

as the total setup cost given the order quantities $\boldsymbol{z} \in \mathbb{R}_+^n$, where $K_0, K_1, \dots, K_n$ are non-negative constants, K_0 is the joint setup cost of all products, and K_i is the individual setup cost of product i ($i = 1, \dots, n$). Hereafter, we refer to equation (1) as *the generalized indicator function*.

The variable ordering cost is proportional to the order quantity for each product. Let c_i be the unit ordering cost for each product i. Then, the variable ordering cost given the order quantities $\boldsymbol{z}$ is $c(\boldsymbol{z}) = \sum_{i=1}^{n} c_i z_i$. Hence, the total ordering cost under the ordering decisions $\boldsymbol{z}$ is

$$\mathcal{K}(\boldsymbol{z}) + c(\boldsymbol{z}) = K_0 \delta\left(\sum_{i=1}^{n} z_i\right) + \sum_{i=1}^{n} K_i \delta(z_i) + \sum_{i=1}^{n} c_i z_i.$$

In addition, we also have the inventory cost due to the mismatch of demand and supply. Specifically, given the order-up-to level y_i, the expected backlogged cost is $\mathrm{E}b_i([D_{i,t} - y_i]^+)$, and the expected holding cost is $\mathrm{E}h_i([y_i - D_{i,t}]^+)$ (noting that $[x]^+ = \max\{0, x\}$), where $h_i(\cdot)$ and $b_i(\cdot)$ are strictly increasing functions. Then, the expected inventory cost for all products in period t given the post-ordering inventory levels $\boldsymbol{y}$ is

$$L_t(\boldsymbol{y}) = \sum_{i=1}^{n} L_{i,t}(y_i) = \sum_{i=1}^{n} \mathrm{E}\left[h_i([y_i - D_{i,t}]^+) + b_i([y_i - D_{i,t}]^-)\right].$$

where $L_{i,t}(y_i) = \mathrm{E}[h_i([y_i - D_{i,t}]^+) + b_i([y_i - D_{i,t}]^-)]$.

The objective is to identify the optimal procurement strategy in each period to minimize the total expected discounted cost throughout the planning horizon. Let $\alpha \in (0, 1]$ be the discount factor and $\tilde{v}_t(\boldsymbol{x})$ be the total expected discounted profit from period t and onward with the starting inventory state $\boldsymbol{x}_t$. Then, for $t = 1, \dots, T$, the dynamic recursion of $\tilde{v}_t(\boldsymbol{x})$ can be formulated as follows:

$$\tilde{v}_t(\boldsymbol{x}) = -\sum_{i=1}^{n} c_i x_i + \min_{\boldsymbol{y} \geq \boldsymbol{x}} \left\{ \widetilde{G}_t(\boldsymbol{y}) + \mathcal{K}(\boldsymbol{y} - \boldsymbol{x}) \right\}, \tag{2}$$

where $\widetilde{G}_t(\boldsymbol{y}) = \sum_{i=1}^{n} [c_i y_i + L_{i,t}(y_i)] + \alpha \mathrm{E}[\tilde{v}_{t+1}(\boldsymbol{y} - \boldsymbol{D}_t)]$. Note that the terminal period is indexed by $T+1$. Following Veinott (1966) and Kalin (1980), we set $\tilde{v}_{T+1}(\boldsymbol{x}) = -\sum_{i=1}^{n} c_i x_i$.

We transform the dynamic recursions in equation (2) as follows to facilitate the subsequent analysis. Define

$$v_t(\boldsymbol{x}) = \tilde{v}_t(\boldsymbol{x}) + \sum_{i=1}^{n} c_i x_i, \tag{3}$$

$$g_{i,t}(y_i) = (1 - \alpha) c_i y_i + L_{i,t}(y_i) + \alpha c_i d_{i,t}, \tag{4}$$

$$g_t(\boldsymbol{y}) = \sum_{i=1}^{n} g_{i,t}(y_i). \tag{5}$$

Then, the dynamic recursions in equation (2) can be rewritten as follows:

$$v_t(\boldsymbol{x}) = \min_{\boldsymbol{y} \geq \boldsymbol{x}} \left\{ G_t(\boldsymbol{y}) + \mathcal{K}(\boldsymbol{y} - \boldsymbol{x}) \right\}, \tag{6}$$

where

$$G_t(\boldsymbol{y}) = g_t(\boldsymbol{y}) + \alpha \mathrm{E}[v_{t+1}(\boldsymbol{y} - \boldsymbol{D}_t)], \tag{7}$$

and $v_{T+1}(\boldsymbol{x}) = 0$ for any $\boldsymbol{x} \in \mathbb{R}^n$.

In addition, we impose the following assumptions.

ASSUMPTION 1. *For all $t = 1, \dots, T$ and $i = 1, \dots, n$, the function $g_{i,t}(y_i)$, defined in equation (4), is continuous and unimodal in y_i with $\lim_{y_i \to -\infty} g_{i,t}(y_i) = \lim_{y_i \to \infty} g_{i,t}(y_i) = \infty$.*

This assumption implies that $g_t(\boldsymbol{y})$ is component-wise unimodal and continuous in $\boldsymbol{y}$. Note that the common assumption on the convexity of the single-period inventory cost is a special case of Assumption 1, for example, when there is a linear backordering cost and a linear holding cost. Due to Assumption 1, there exists a global minimizer $\boldsymbol{\eta}_t$ such that

$$\boldsymbol{\eta}_t = \arg\min_{\boldsymbol{y}} g_t(\boldsymbol{y}).$$

REMARK 1. *Note that different demand correlations of the n products potentially lead to different global minima $\boldsymbol{\eta}_t$ under the same marginal contribution of each demand.*

We also impose the following assumption on $\boldsymbol{\eta}_t$.

ASSUMPTION 2. *For $t = 1, \dots, T-1$, we assume $\boldsymbol{\eta}_t \leq \boldsymbol{\eta}_{t+1}$.*

This assumption resembles the counterparts by Veinott (1966) and Kalin (1980) for the same purpose. One can refer to Kalin (1980) for a more detailed discussion. For instance, if the demands, holding costs, and backordering costs are stationary, the above assumption is naturally satisfied. Under linear backordering costs and linear holding costs, Assumption 2 can be satisfied if demands are stochastically increasing over time. In practice, it is common to amortize the inventory cost into the backlogged and holding cost per unit of product which can be viewed as linear costs. Stochastic increasing demands are also common, for example, stationary demands. For example, the market size of daily necessities is often stable or estimated to increase steadily.

REMARK 2 (Demands With Customer Choice). *In many retail settings, the demands for different products typically result from customer choice. Suppose that the selling prices of the n products are* $\boldsymbol{p}_t = (p_1, \dots, p_n)$. *Let* D_t *be the total demand for the n products, and* $D_{i,t}(\boldsymbol{p})$ *be the demand for product i* ($i = 1, \dots, n$) *in period t such that* $D_t = \sum_{i=1}^n D_{i,t}(\boldsymbol{p})$. *Typically, we assume that* $D_{i,t}(\boldsymbol{p})$ *is stochastically decreasing in* p_i *but increasing in* $p_j, i \neq j$. *For example, under a multinomial choice model, we can write*

$$D_{i,t}(\boldsymbol{p}) = D_t \frac{f_i(p_i)}{\sum_{j=1}^n f_j(p_j)},$$

where $f_j(p_j) = e^{u_j - \beta_j p_j}$ *for some constant* u_j, β_j *for* $j = 1, \dots, n$ *(Gallego and Wang, 2014). In this case, the demands for different products are negatively correlated. However, because the selling prices are fixed in each period, we can omit the dependency of demands on prices. In this case, the single-period expected cost in period t under the order-up-to levels* $\boldsymbol{y}$ *is* $\sum_{i=1}^n L_{i,t}(y_i) - \sum_{i=1}^n p_i d_i$. *It is clear that the term* $\sum_{i=1}^n p_i d_i$ *is independent of the ordering decisions. Thus, this term does not affect our results and can be omitted in our analysis.*

4 Preliminaries

In this section, we first introduce definitions and preliminary results that are useful in our subsequent analysis, and particularly in the characterization of the optimal policy given in Section 5. First, we present the results on the subadditivity of the generalized indicator function $\mathcal{K}$ and the properties of $(\mathcal{K}, \boldsymbol{\eta})$-quasi-convexity, then we introduce the definition of the $(\sigma, \omega, \boldsymbol{S})$ policy.

We define our notation as follows. For any $\boldsymbol{x}', \boldsymbol{x}'' \in \mathbb{R}^n$, let

$\mathbb{N}$	:=	$\{1, \dots, n\}$;
$\boldsymbol{x}' \vee \boldsymbol{x}''$	:=	the component-wise maximum vector of vectors $\boldsymbol{x}'$ and $\boldsymbol{x}''$, that is ,$\boldsymbol{x} = \boldsymbol{x}' \vee \boldsymbol{x}''$, if $x_i = \max\{x_i', x_i''\}$ for all i;
$\boldsymbol{x}' \leq \boldsymbol{x}''$	:=	$\boldsymbol{x}'$ is component-wise less than $\boldsymbol{x}''$, *that is*, $x_i' \leq x_i''$ for all i;
$\boldsymbol{x}' < \boldsymbol{x}''$	:=	$\boldsymbol{x}'$ is strictly less than $\boldsymbol{x}''$, if $\boldsymbol{x}' \leq \boldsymbol{x}''$ and $\boldsymbol{x}' \neq \boldsymbol{x}''$;
$\boldsymbol{x}' <_{\ell} \boldsymbol{x}''$	:=	$x_i' < x_i''$, for $\forall i \in \ell \subseteq \mathbb{N}$, and $x_i' = x_i''$, for $\forall i \in \mathbb{N} \setminus \ell$.

Clearly, $\boldsymbol{x}' \leq_{\emptyset} \boldsymbol{x}$ implies that $\boldsymbol{x}' = \boldsymbol{x}$. Therefore, in the subsequent analysis, we rule out the case $\ell = \emptyset$ or $\rho = \emptyset$ whenever $\rho \subseteq \ell$.

The following lemma shows that the generalized indicator function $\mathcal{K}$ is a subadditive function.

LEMMA 1. *The generalized indicator function* $\mathcal{K}$ *in equation (1) is subadditive, that is,*

$$\mathcal{K}(z' + z'') \leq \mathcal{K}(z') + \mathcal{K}(z''), \forall z', z'' \in \mathbb{R}_+^n.$$

We next propose the notions of $\mathcal{K}$-nondecreasing and $(\mathcal{K}, \boldsymbol{\eta})$-quasi-convexity.

DEFINITION 1 ($\mathcal{K}$-nondecreasing). *Let* $\mathcal{K}$ *be the generalized indicator function, we say that a function* $f : \mathbb{R}^n \to \mathbb{R}$ *is* $\mathcal{K}$*-nondecreasing on a convex domain* $\mathbb{X} \subseteq \mathbb{R}^n$ *if*

$$f(\boldsymbol{x}) \leq f(\boldsymbol{y}) + \mathcal{K}(\boldsymbol{y} - \boldsymbol{x}), \ \forall \boldsymbol{x} \leq \boldsymbol{y}, \ \boldsymbol{x}, \boldsymbol{y} \in \mathbb{X}.$$

DEFINITION 2 ($(\mathcal{K}, \boldsymbol{\eta})$-quasi-convexity). *Let* $\boldsymbol{\eta} \in \mathbb{X} \subset \mathbb{R}^n$ *and* $\mathcal{K}$ *be the generalized indicator function, we say that a function* $f : \mathbb{R}^n \to \mathbb{R}$ *is* $(\mathcal{K}, \boldsymbol{\eta})$*-quasi-convex on* $\mathbb{X}$ *if the following conditions hold for* $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{X}$*:*

(1) $f(\boldsymbol{x}) \leq f(\boldsymbol{y}) + \mathcal{K}(\boldsymbol{y} - \boldsymbol{x})$ *for* $\boldsymbol{\eta} \leq \boldsymbol{x} \leq \boldsymbol{y}$;
(2) $f(\boldsymbol{x}) \geq f(\boldsymbol{y})$ *for* $\boldsymbol{x} \leq \boldsymbol{y} \leq \boldsymbol{x} \vee \boldsymbol{\eta}$.

These two definitions are critical to show the optimality of the $(\sigma, \omega, \boldsymbol{S})$ policy.

REMARK 3. *It is clear that these definitions are the generalizations of K-nondecreasing functions and* $(K, \boldsymbol{\eta})$*-quasi-convex functions by Kalin (1980), respectively. The key difference is that K is a nonnegative constant by Kalin (1980), while it is a generalized indicator function in our case. Our conceptualization of the* $(\mathcal{K}, \boldsymbol{\eta})$*-quasi-convexity is necessary for our setting, since Kalin (1980) only considers the case with a joint setup cost* $K_0\delta(\sum_{i=1}^n z_i)$ *but no individual setup costs, whereas we consider not only a joint setup cost for all products but also an individual setup cost for each product. Moreover, we allow* $\mathcal{K}$ *to be a general subadditive function (see Section 7.2 for more details).*

The following lemma summarizes the properties of $(\mathcal{K}, \boldsymbol{\eta})$-quasi-convexity.

LEMMA 2. *We have the following properties for* $(\mathcal{K}, \boldsymbol{\eta})$*-quasi-convexity:*

(1) *A convex function* $f(x)$ *on* $\mathbb{X} \subset \mathbb{R}$ *is also a* $(0, \eta)$*-quasi-convex function with* $\mathcal{K} = 0$ *for all* $x \in \mathbb{X}$ *and* $\eta = \arg\min_{x \in \mathbb{R}} f(x)$.
(2) *Let* $f(\boldsymbol{x})$ *be* $(\mathcal{K}_1, \boldsymbol{\eta})$*-quasi-convex on* $\mathbb{X} \subset \mathbb{R}^n$. *If* $\mathcal{K}_2(z) \geq \mathcal{K}_1(z)$ *for all* $z \in \mathbb{R}_+^n$, *then* $f(\boldsymbol{x})$ *is also* $(\mathcal{K}_2, \boldsymbol{\eta})$*-quasi-convex on* $\mathbb{X}$.

(3) *If* $f_1(\boldsymbol{x})$ *is* $(\boldsymbol{\mathcal{K}}_1, \boldsymbol{\eta})$*-quasi-convex and* $f_2(\boldsymbol{x})$ *is* $(\boldsymbol{\mathcal{K}}_2, \boldsymbol{\eta})$*-quasi-convex on* $\mathbb{X} \subset \mathbb{R}^n$*, then* $\lambda_1 f_1(\boldsymbol{x}) + \lambda_2 f_2(\boldsymbol{x})$ *is* $(\lambda_1 \boldsymbol{\mathcal{K}}_1 + \lambda_2 \boldsymbol{\mathcal{K}}_2, \boldsymbol{\eta})$*-quasi-convex on* $\mathbb{X}$ *for any* $\lambda_1, \lambda_2 \geq 0$.

(4) *Suppose* $f_i(x_i)$ *is* (K_i, η_i)*-quasi-convex on* $\mathbb{X}_i \subset \mathbb{R}$ *for* $i = 1, \ldots, n$*, that is,* $f_i(x_i) \geq f_i(y_i)$ *for* $x_i < y_i \leq x_i \vee \eta_i$ *and* $f_i(x_i) \leq f_i(y_i) + K_i \delta(y_i - x_i)$ *for* $\eta_i \leq x_i < y_i$*. Then* $f(\boldsymbol{x}) = \sum_{i=1}^n f_i(x_i)$ *is* $(\boldsymbol{\mathcal{K}}, \boldsymbol{\eta})$*-quasi-convex on* $\mathbb{X} = \prod_{i=1}^n \mathbb{X}_i \subset \mathbb{R}^n$ *with* $\boldsymbol{\eta} = (\eta_1, \ldots, \eta_n)$.

(5) *A component-wise convex function* $f(\boldsymbol{x}) = \sum_{i=1}^n f_i(x_i)$ *is* $(0, \boldsymbol{\eta})$*-quasi-convex on* $\mathbb{X} \subset \mathbb{R}^n$ *if* $\boldsymbol{\eta} \in \arg\max_{x \in \mathbb{X}} f(\boldsymbol{x})$*, where* 0 *denotes* $\boldsymbol{\mathcal{K}}(z) = 0$ *for all* $z \in \mathbb{R}_+^n$*, that is,* $K_0 = K_1 = \cdots = K_n = 0$.

It is well-known that the monotonicity properties of an ordering policy are important to characterize the optimal policy, that is, if it is optimal to order in a particular state, then it must be optimal to order in a (component-wise) smaller state. The following lemma presents another important property of $(\boldsymbol{\mathcal{K}}, \boldsymbol{\eta})$-quasi-convexity that ensures the ordering monotonicity.

LEMMA 3. *Suppose that function* $f(\boldsymbol{x})$ *is* $(\boldsymbol{\mathcal{K}}, \boldsymbol{\eta})$*-quasi-convex on* $\mathbb{X} \subset \mathbb{R}^n$*. Define function* $\bar{f}(\boldsymbol{x}) = \min_{y \geq x}\{f(\boldsymbol{y}) + \boldsymbol{\mathcal{K}}(\boldsymbol{y} - \boldsymbol{x})\}$ *and let* $\boldsymbol{y}^*(\boldsymbol{x}) = \arg\min_{y \geq x}[f(\boldsymbol{y}) + \boldsymbol{\mathcal{K}}(\boldsymbol{y} - \boldsymbol{x})]$*. Then for any* $\boldsymbol{x}'' \leq \boldsymbol{x}' \leq \boldsymbol{x}'' \vee \boldsymbol{\eta}$*, if* $\boldsymbol{x}'' <_\rho \boldsymbol{x}' <_\ell \boldsymbol{y}^*(\boldsymbol{x}')$ *such that* $\rho \subseteq \ell$*, we have* $\boldsymbol{y}^*(\boldsymbol{x}'') = \boldsymbol{y}^*(\boldsymbol{x}')$.

REMARK 4. *To understand the implications of Lemma 3, consider an inventory problem with the optimal value function* $\bar{f}(\boldsymbol{x})$ *and the optimal order-up-to levels defined as* $\boldsymbol{y}^*(\boldsymbol{x})$ *under the state* $\boldsymbol{x}$*. Then, Lemma 3 states that if it is optimal to order positive quantities for a set of products under the state* $\boldsymbol{x}'$*, then it is also optimal to order some positive quantities for the state* $\boldsymbol{x}''$ *which satisfies* $\boldsymbol{x}'' <_\rho \boldsymbol{x}'$ *with* $\rho \subseteq \ell$*. Furthermore, the order-up-to levels for* $\boldsymbol{x}'$ *and* $\boldsymbol{x}''$ *are the same. This lemma also implies that we can find series of reorder sets and reorder points, such that for states less than some reorder point for products in the corresponding set, it is optimal to bring these products up to the same inventory levels. We refer to this property as the ordering monotonicity property for such states* $\boldsymbol{x}'$ *and* $\boldsymbol{x}''$*, which can substantially facilitate our analysis.*

Finally, we define the $(\sigma, \omega, \boldsymbol{S})$ policy that characterizes the structure of the optimal policy for our problem.

DEFINITION 3 ($(\sigma, \omega, \boldsymbol{S})$ Policy). *Let* $\boldsymbol{S} \in \mathbb{R}^n$ *be the vector of base stock levels, and let* σ, ω *be two disjoint regions in* $\mathbb{R}^n$*. Let* $\boldsymbol{\eta} \in \mathbb{R}^n$ *be the vector of base stock levels that minimizes the single-period inventory cost. A procurement policy is called a* $(\sigma, \omega, \boldsymbol{S})$ *policy, if the order-up-to inventory levels, denoted by* $\boldsymbol{y}(\boldsymbol{x})$ *for the net inventory vector* $\boldsymbol{x}$ *(prior to ordering), have the following properties:*

(1) *For* $\boldsymbol{x} \in \sigma$*, order up to* $\boldsymbol{S}$*, that is,* $\boldsymbol{y}(\boldsymbol{x}) = \boldsymbol{S}$*. For each index set* $\ell \subseteq N$*, there exists a set of vectors* $\boldsymbol{s}_1^\ell, \boldsymbol{s}_2^\ell, \ldots, \boldsymbol{s}_{k^\ell}^\ell$*, where* k^ℓ *may be infinite. Define* $\sigma^\ell = \{\boldsymbol{x} | \boldsymbol{x} <_\rho \boldsymbol{s}_i^\ell$ *for some* $i = 1, \ldots, k^\ell$ *and* $\rho \subseteq \ell \setminus \emptyset\}$.

(2) *For* $\boldsymbol{x} \in \omega$*, order nothing, that is,* $\boldsymbol{y}(\boldsymbol{x}) = \boldsymbol{x}$*. Furthermore, if* $\boldsymbol{x} \in \omega$ *and* $\boldsymbol{x}' \geq \boldsymbol{x}$*, then* $\boldsymbol{x}' \in \omega$*. In addition,* $\{\boldsymbol{x} | \boldsymbol{x} \geq \boldsymbol{\eta}\} \subseteq \omega$.

(3) *For* $\boldsymbol{x} \notin \sigma \cup \omega$*, order certain amounts of certain products, that is,* $\boldsymbol{y}(\boldsymbol{x}) > \boldsymbol{x}$.

(4) *Furthermore, for* $\boldsymbol{x}' \leq \boldsymbol{x} \leq \boldsymbol{x}' \vee \boldsymbol{\eta}$*, if* $\boldsymbol{x}' <_\rho \boldsymbol{x} <_\ell \boldsymbol{y}(\boldsymbol{x})$ *with* $\rho \subseteq \ell$*, then* $\boldsymbol{y}(\boldsymbol{x}') = \boldsymbol{y}(\boldsymbol{x})$.

Properties (1)–(3) indicate how to order for states in σ, ω and for state in neither σ nor ω. Property (4) appears complex at the first glance. Note that Property (4) in Definition 3 is a direct implication of Lemma 3. To understand this property more intuitively, recall the classical one-product (s, S) policy that when the inventory $x < s$ we order up to S. In other words, if we already know that $y(x) = S$, the (s, S) policy implies that we should also order up to S for any state $x' < x$, that is, $y(x') = S = y(x)$. Property (4) just generalizes such idea to the multiple product setting since the inventory state is multi-dimensional, with the help of the partial order $<_\ell$ and $<_\rho$.

REMARK 5. *Our definition is consistent with Johnson (1967), who also uses the notation* σ *to denote the reorder set (i.e., the set of inventory state for which an order should be placed). Johnson (1967) shows that a stationary* $(\sigma, \boldsymbol{S})$ *policy is optimal for an infinite-horizon multiple-product system with only a joint setup cost: order up to* $\boldsymbol{S}$ *for all* $\boldsymbol{x} \in \sigma$*, and otherwise, order nothing. However, Johnson (1967) does not explicitly characterize the structure of the reorder set* σ.

For the benchmarking purpose, we also provide the definition of the $(\sigma, \boldsymbol{S})$ policy introduced by Kalin (1980), that is, do not order if the inventory state belongs to σ, otherwise order up to $\boldsymbol{S}$. To be consistent with our notation, we change the symbol σ by Kalin (1980) to ω. In the subsequent analysis, we refer to the policy by Kalin (1980) as the $(\omega, \boldsymbol{S})$ policy.

DEFINITION 4 ($(\omega, \boldsymbol{S})$ policy (Kalin, 1980)). *Let* $\boldsymbol{S} \in \mathbb{R}^n$ *be the vector of base stock levels, and* ω *be a subset in* $\mathbb{R}^n$*. A procurement policy is called an* $(\omega, \boldsymbol{S})$ *policy, if the order-up-to inventory levels, denoted by* $\boldsymbol{y}(\boldsymbol{x})$ *for the net inventory state (or vector)* $\boldsymbol{x}$ *(prior to ordering), has the following properties:*

(1) *For* $\boldsymbol{x} \in \omega$*, do not order, that is,* $\boldsymbol{y}(\boldsymbol{x}) = \boldsymbol{x}$ *and* ω *is the no-order set.*

(2) *For* $\boldsymbol{x} \leq \boldsymbol{S}, \boldsymbol{x} \notin \omega$*, order up to* $\boldsymbol{S}$*, that is,* $\boldsymbol{y}(\boldsymbol{x}) = \boldsymbol{S}$.

(3) *If* $\boldsymbol{x} \in \omega$ *and* $\boldsymbol{x} \leq^{\eta} \boldsymbol{y}$*, then* $\boldsymbol{y} \in \omega$ *(noting that* $\leq^{\eta}$ *is a partial order on a subset of* $\mathbb{R}^n$ *if either of the following conditions hold: (i)* $\boldsymbol{x} \vee \boldsymbol{\eta} \leq \boldsymbol{y}$ *or (ii)* $\boldsymbol{x} \leq \boldsymbol{y} \leq \boldsymbol{x} \vee \boldsymbol{y}$*, where* $\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{\eta}$ *are vectors of the subset.)*

We summarize the differences between the $(\sigma, \omega, \boldsymbol{S})$ policy and the $(\omega, \boldsymbol{S})$ policy as follows:

(i) Kalin (1980) characterizes the $(\omega, \boldsymbol{S})$ policy by introducing the partial order $\leq^{\boldsymbol{\eta}}$ defined on the no-order set ω based on the partial order. By contrast, for the $(\sigma, \omega, \boldsymbol{S})$ policy, we define the reorder set σ based on a new partial order with respect to an index set $\ell \subset \mathbb{N}$. This definition and the new partial order allow us to derive more refined properties of the ordering policy, in particular the regions in which it is optimal not to order up to $\boldsymbol{S}$.

(ii) In an $(\omega, \boldsymbol{S})$ policy, one should either order up to $\boldsymbol{S}$, or order nothing, for any state in $\{\boldsymbol{x}|\boldsymbol{x} \leq \boldsymbol{S}\}$. However, it is no longer true for the $(\sigma, \omega, \boldsymbol{S})$ policy because of the presence of individual setup costs: there are some states in $\{\boldsymbol{x}|\boldsymbol{x} \leq \boldsymbol{S}\}$ for which one should order but not necessarily to $\boldsymbol{S}$. In fact, these individual setup costs impose new technical challenges to show the optimality of the $(\sigma, \omega, \boldsymbol{S})$ policy for our problem.

(iii) Our ordering monotonicity property also applies for those states that should order but not up to $\boldsymbol{S}$, that is, states that are neither in σ nor in ω, while Kalin (1980) does not provide any characterization for those states.

(iv) Finally, the $(\omega, \boldsymbol{S})$ policy is driven only by the joint setup cost K_0 while the $(\sigma, \omega, \boldsymbol{S})$ policy is driven by both the joint setup cost and the individual setup costs.

As a result, the $(\omega, \boldsymbol{S})$ policy can be viewed as a special case of the $(\sigma, \omega, \boldsymbol{S})$ policy; see discussions in Remark 7.

Note that for a $(\sigma, \omega, \boldsymbol{S})$ policy, the ordering behavior is fairly simple for the regions σ and ω, but it is not intuitive outside of these two regions, that is, the region under which it is optimal not to order up to $\boldsymbol{S}$. Moreover, the boundaries of regions σ and ω are typically determined by nonlinear functions. To better understand the optimal policy, we derive explicit bounds for the parameters of the optimal policy, including the order-up-to-levels and the boundaries of sets σ (see Section 6.1) and $\theta \setminus \sigma$ (see the index-based bounds in Appendix F in the E-Companion). We also derive a lower bound system which is asymptotically optimal as the coefficient of variation of demand decreases to zero. The lower bound system is indeed a multi-product lot-sizing problem which is well-studied in the literature and can be effectively solved (see Section 6.2). Motivated by the optimal policy structure and bounds, we design simple and effective heuristics to approximate the optimal policy for $\boldsymbol{x} \notin \sigma \cup \omega$ (see Section 7).

5 The Optimality of the (σ, ω, S) Policy

In this section, we exploit the $(\mathcal{K}, \boldsymbol{\eta})$-quasi-convexity to show that the optimal procurement policy is a $(\sigma, \omega, \boldsymbol{S})$ policy for each period. For notational simplicity, let $H_t(\boldsymbol{y}) = \mathrm{E}[v_{t+1}(\boldsymbol{y} - \boldsymbol{D}_t)]$. Then we have $G_t(\boldsymbol{y}) = g_t(\boldsymbol{y}) + \alpha H_t(\boldsymbol{y})$. Define $J_t(\boldsymbol{x}) = \min_{\boldsymbol{y} > \boldsymbol{x}}\{G_t(\boldsymbol{y}) + \mathcal{K}(\boldsymbol{y} - \boldsymbol{x})\}$, which is the expected discounted total cost in period t if an optimal positive order is placed.

Then, the dynamic recursions in equation (6) can be reformulated as follows:

$$v_t(\boldsymbol{x}) = \min\{G_t(\boldsymbol{x}), J_t(\boldsymbol{x})\},$$

where $G_t(\boldsymbol{x})$ is the expected discounted total profit in period t if no order is placed.

To facilitate our subsequent analysis, we introduce the following terms to characterize the structure of the optimal policy. Let $\boldsymbol{y}_t^*(\boldsymbol{x}) = \arg\min_{\boldsymbol{y} \geq \boldsymbol{x}}\{G_t(\boldsymbol{y}) + \mathcal{K}(\boldsymbol{y} - \boldsymbol{x})\}$ be the optimal order-up-to levels given the inventory state $\boldsymbol{x}$. Define

$$\theta_t = \{\boldsymbol{x}|G_t(\boldsymbol{x}) > J_t(\boldsymbol{x})\}, \quad (8)$$

$$\omega_t = \{\boldsymbol{x}|G_t(\boldsymbol{x}) \leq J_t(\boldsymbol{x})\}. \quad (9)$$

Sets θ_t and ω_t can be interpreted as the reorder set and the no-order set in period t, respectively.

Next, we provide some partial characterizations of θ_t and ω_t. Let $\boldsymbol{S}_t$ be a global minimum of $G_t(\boldsymbol{y})$ defined in equation (7), that is,

$$\boldsymbol{S}_t \in \arg\min_{\boldsymbol{y}} G_t(\boldsymbol{y}). \quad (10)$$

Note that if $G_t(\boldsymbol{y})$ has multiple global minimizers, one can choose any one of them, for example, in a lexicographical order.

For the given $\boldsymbol{S}_t$, there exists a curve, that is, a set of vectors (reorder points) $\boldsymbol{s}_t^{\ell} \in \mathbb{R}^n$ with $\boldsymbol{s}_t^{\ell} <_{\ell} \boldsymbol{S}_t$, with respect to some set $\ell \subseteq \mathbb{N} \backslash \emptyset$ such that

$$v_t(\boldsymbol{s}_t^{\ell}) = G_t(\boldsymbol{S}_t) + \mathcal{K}(\boldsymbol{S}_t - \boldsymbol{s}_t^{\ell}), \quad (11)$$

where $\mathcal{K}(\boldsymbol{S}_t - \boldsymbol{s}_t^{\ell}) = K_0 + \sum_{i \in \ell} K_i$. Let $\mathcal{R}_t^{\ell}$ be the set of reorder points which are the solutions to equation (11). Moreover, we say that a vector $\boldsymbol{s}_t \in \mathcal{R}_t^{\ell}$ is on the frontier if we cannot find another $\boldsymbol{s}_t' \in R_t^{\ell}$ such that $\boldsymbol{s}_t <_{\rho} \boldsymbol{s}_t'$ for some $\rho \subseteq \ell \setminus \emptyset$. Let

$$\mathcal{F}_t^{\ell} = \{\boldsymbol{s}_{1,t}^{\ell}, \ldots, \boldsymbol{s}_{k^{\ell},t}^{\ell}\},$$

where k^{ℓ} may be infinite, be the set of all reorder points on the frontier.

We define

$$\sigma_t^{\ell} = \Big\{\boldsymbol{x}|\boldsymbol{x} <_{\rho} \boldsymbol{s}_t^{\ell} \text{ for some } \rho \subseteq \ell \setminus \emptyset \text{ and } \boldsymbol{s}_t^{\ell} \in \mathcal{F}_t^{\ell}, \quad \text{i.e., } v_t(\boldsymbol{s}_t^{\ell}) = G_t(\boldsymbol{S}_t) + \mathcal{K}(\boldsymbol{S}_t - \boldsymbol{s}_t^{\ell})\Big\}. \quad (12)$$

In this definition, $\boldsymbol{s}_t^{\ell}$ is on the frontier where it is optimal to order the products in set ℓ and bring the inventory levels to $\boldsymbol{S}_t$, and σ_t^{ℓ} is the region in which it is optimal to order some or all of products in set ℓ. Define

$$\sigma_t = \Big\{\boldsymbol{x}|\boldsymbol{x} \in \sigma_t^{\ell} \text{ for some } \ell \subseteq \mathbb{N} \setminus \emptyset\Big\} = \cup_{\ell \subseteq \mathbb{N} \backslash \emptyset} \sigma_t^{\ell}, \quad (13)$$

that is, σ_t represents the union of all possible order-up-to-$\boldsymbol{S}_t$ sets. Furthermore, it is clear that $\sigma_t^{\ell} \subseteq \sigma_t \subseteq \theta_t$, that is, σ_t^{ℓ} is a subset of the states where it is optimal to order up to $\boldsymbol{S}_t$, and all sets in σ_t (where it is optimal to order up to $\boldsymbol{S}_t$) are just subsets of the reorder set θ_t (where it is optimal to order for certain products up to certain inventory levels but not necessarily up to $\boldsymbol{S}_t$).

REMARK 6. *(i) Note that σ_t is the union of σ_t^{ℓ} for all $\ell \subseteq \mathbb{N}\backslash\emptyset$, where the latter is further defined on the reorder frontier set $\mathcal{F}_t$ and the partial order $<_\rho$ with $\rho \subseteq \ell, \rho \neq \emptyset$. To better understand the definition of σ_t, we further discuss the interrelationship among the partial order $<_\rho$, the reorder frontier $\mathcal{F}_t^{\ell}$, and $\mathcal{R}_t^{\ell}$. Note that the reorder point set $\mathcal{R}_t^{\ell}$ in general is an area. However, by exploiting the monotonicity property in Lemma 3, we can refine $\mathcal{R}_t^{\ell}$ by a frontier set $\mathcal{F}_t^{\ell}$. For example in Section 5.1, the reorder frontier set for $\rho = \ell = \{1, 2\}$ consists of only six points. (ii) The partial order we defined is novel in the existing literature; such partial order explicitly characterizes the reorder sets, including those inventory states that should be ordered up to $\boldsymbol{S}_t$ and other inventory states that should be ordered but not necessarily to $\boldsymbol{S}_t$. By contrast, the partial order $<^{\boldsymbol{\eta}}$ in Kalin (1980) applies to the no-order set. (iii) Furthermore, it should be aware that, it may not be optimal to order up to $\boldsymbol{S}_t$ for $\boldsymbol{x} \le \boldsymbol{s}^{\ell}$ for any given $\boldsymbol{s}^{\ell}$; see examples in Section 5.1 for the case when $\boldsymbol{s}^{\{1\}} = (-1, 9)$.*

We next proceed to show the optimality of a $(\sigma_t, \omega_t, \boldsymbol{S}_t)$ policy by several inductive steps: assuming that $v_{t+1}(\boldsymbol{x})$ is $(\boldsymbol{\mathcal{K}}, \boldsymbol{\eta}_{t+1})$-quasi-convex, $\boldsymbol{\mathcal{K}}$-nondecreasing and continuous, we sequentially show

(i) The continuity and the $(\boldsymbol{\mathcal{K}}, \boldsymbol{\eta}_t)$-quasi-convexity of $G_t(\boldsymbol{y})$ (Lemma 4).
(ii) the optimality of a $(\sigma_t, \omega_t, \boldsymbol{S}_t)$ inventory policy for the dynamic program in equation (6) and the properties of that inventory policy (Lemma 5).
(iii) Finally, we show that $v_t(\boldsymbol{x})$ is indeed $(\boldsymbol{\mathcal{K}}, \boldsymbol{\eta}_t)$-quasi-convex, $\boldsymbol{\mathcal{K}}$-nondecreasing and continuous (Lemma 6), and the optimality of a $(\sigma_t, \omega_t, \boldsymbol{S}_t)$ policy (Theorem 1), by induction.

We first show the $(\boldsymbol{\mathcal{K}}, \boldsymbol{\eta}_t)$-quasi-convexity of $G_t(\boldsymbol{y})$.

LEMMA 4. *If $v_{t+1}(\boldsymbol{x})$ is $(\boldsymbol{\mathcal{K}}, \boldsymbol{\eta}_{t+1})$-quasi-convex, $\boldsymbol{\mathcal{K}}$-nondecreasing and continuous, then we have*

(1) *$H_t(\boldsymbol{y})$ is $(\boldsymbol{\mathcal{K}}, \boldsymbol{\eta}_t)$-quasi-convex, $\boldsymbol{\mathcal{K}}$-nondecreasing and continuous.*
(2) *$G_t(\boldsymbol{y})$ is $(\boldsymbol{\mathcal{K}}, \boldsymbol{\eta}_t)$-quasi-convex and continuous.*
(3) *$\{\boldsymbol{x} | \boldsymbol{x} \ge \boldsymbol{\eta}_t\} \subseteq \omega_t$.*

With the $(\boldsymbol{\mathcal{K}}, \boldsymbol{\eta}_t)$-quasi-concavity of $G_t(\boldsymbol{y})$ in $\boldsymbol{y}$, we can show that a $(\sigma_t, \omega_t, \boldsymbol{S}_t)$ policy is optimal for the dynamic program in equation (6), and we obtain the following properties of the $(\sigma_t, \omega_t, \boldsymbol{S}_t)$ policy.

LEMMA 5. *Suppose that $G_t(\boldsymbol{y}_t)$ is a $(\boldsymbol{\mathcal{K}}, \boldsymbol{\eta}_t)$-quasi-convex function. Then, for the dynamic program in equation (6), the optimal procurement strategy is a $(\sigma_t, \omega_t, \boldsymbol{S}_t)$ policy. In addition, the $(\sigma_t, \omega_t, \boldsymbol{S}_t)$ policy has the following properties:*

(1) *For $\boldsymbol{x}'' \le \boldsymbol{x}' \le \boldsymbol{x}'' \vee \boldsymbol{\eta}_t$, if $\boldsymbol{x}'' <_\rho \boldsymbol{x}' <_\ell \boldsymbol{y}_t^*(\boldsymbol{x}')$ with $\rho \subseteq \ell$, then $\boldsymbol{y}_t^*(\boldsymbol{x}'') = \boldsymbol{y}_t^*(\boldsymbol{x}')$.*
(2) *$\boldsymbol{S}_t \ge \boldsymbol{\eta}_t$ and $\boldsymbol{S}_t \in \omega_t$.*
(3) *For $\boldsymbol{x} \in \sigma_t^{\ell} \subseteq \sigma_t$, if $\boldsymbol{x}' <_\rho \boldsymbol{x}$ with $\rho \subseteq \ell$, then $\boldsymbol{x}' \in \sigma_t^{\ell}$.*
(4) *For $\boldsymbol{x} \in \omega_t$, if $\boldsymbol{x}' \ge \boldsymbol{\eta}_t$ or $\boldsymbol{x} \le \boldsymbol{x}' \le \boldsymbol{x} \vee \boldsymbol{\eta}_t$, then $\boldsymbol{x}' \in \omega_t$.*

We next show that the continuity, $\boldsymbol{\mathcal{K}}$-nondecreasing and $(\boldsymbol{\mathcal{K}}, \boldsymbol{\eta}_t)$-quasi-convexity of $v_t(\boldsymbol{x})$.

LEMMA 6. *For all $t = 1, \ldots, T+1$, $v_t(\boldsymbol{x})$ is $(\boldsymbol{\mathcal{K}}, \boldsymbol{\eta}_t)$-quasi-convex, $\boldsymbol{\mathcal{K}}$-nondecreasing and continuous.*

With Lemmas 4–6, we are now ready to show the optimality of the $(\sigma, \omega, \boldsymbol{S})$ policy.

THEOREM 1. *For $t = 1, \ldots, T$, the optimal procurement strategy in each period t is a $(\sigma_t, \omega_t, \boldsymbol{S}_t)$ policy: there exist sets ω_t, σ_t which are subsets in $\mathbb{R}^n$ (defined in equations (9) and (13), respectively), and constant vectors $\boldsymbol{S}_t$, reorder frontier sets $\mathcal{F}_t^{\ell}, \ell \subseteq \mathbb{N}$ (defined in equations (10) and (5), respectively), such that $\boldsymbol{s}_t^{\ell} <_\ell \boldsymbol{S}_t$ for any $\boldsymbol{s}_t^{\ell} \in \mathcal{F}_t^{\ell}$ and $\boldsymbol{S}_t \ge \boldsymbol{\eta}_t$, the optimal policy $\boldsymbol{y}_t^*(\boldsymbol{x})$ for each starting inventory state $\boldsymbol{x}$ can be described as follows:*

(1) *For $\boldsymbol{x} \in \sigma_t^{\ell} = \{\boldsymbol{x} | \boldsymbol{x} <_\rho \boldsymbol{s}_t^{\ell}$ for some $\rho \subseteq \ell$ and $\boldsymbol{s}_t^{\ell} \in \mathcal{F}_t^{\ell}\}$, it is optimal to order up to $\boldsymbol{S}_t$ (and only order the products in set ℓ), that is, $\boldsymbol{y}_t^*(\boldsymbol{x}) = \boldsymbol{S}_t$. Furthermore, if $\boldsymbol{x} \in \sigma_t^{\ell}$ and $\boldsymbol{x}' \le_\rho \boldsymbol{x}$ for any $\rho \subseteq \ell$, then $\boldsymbol{x}' \in \sigma_t^{\ell}$.*
(2) *For $\boldsymbol{x} \in \omega_t$, it is optimal to order nothing, that is, $\boldsymbol{y}_t^*(\boldsymbol{x}) = \boldsymbol{x}$. Furthermore, if $\boldsymbol{x} \in \omega_t$ and $\boldsymbol{x}_t' \ge \boldsymbol{x}_t$, then $\boldsymbol{x}' \in \omega_t$. In addition, $\{\boldsymbol{x} | \boldsymbol{x} \ge \boldsymbol{\eta}_t\} \subseteq \omega_t$.*
(3) *For $\boldsymbol{x} \notin \sigma_t \cup \omega_t$, it is optimal to order certain amounts of certain products, that is, $\boldsymbol{y}_t^*(\boldsymbol{x}) > \boldsymbol{x}$.*
(4) *Furthermore, for $\boldsymbol{x}' \le \boldsymbol{x} \le \boldsymbol{x}' \vee \boldsymbol{\eta}_t$, if $\boldsymbol{x}' <_\rho \boldsymbol{x} <_\ell \boldsymbol{y}_t^*(\boldsymbol{x})$ with $\rho \subseteq \ell$, then $\boldsymbol{y}_t^*(\boldsymbol{x}') = \boldsymbol{y}_t^*(\boldsymbol{x})$.*

Theorem 1 shows that the optimal $(\sigma, \omega, \boldsymbol{S})$ policy is optimal for the inventory system with multiple products and multiple setup costs. The optimal $(\sigma, \omega, \boldsymbol{S})$ policy actually generalizes the $(\omega, \boldsymbol{S})$ policy by Kalin (1980) by further considering individual setup costs. To better understand the optimal $(\sigma, \omega, \boldsymbol{S})$ policy and its relationship with the $(\omega, \boldsymbol{S})$ policy by Kalin (1980), we numerically illustrate both policies in the next section.

REMARK 7. *If we set $K_i = 0$ for all i in our setting, the dynamic program and the optimal policy degenerate into those by Kalin (1980). Therefore, the $(\omega, \boldsymbol{S})$ policy by Kalin (1980) is a special case of the $(\sigma, \omega, \boldsymbol{S})$ policy.*

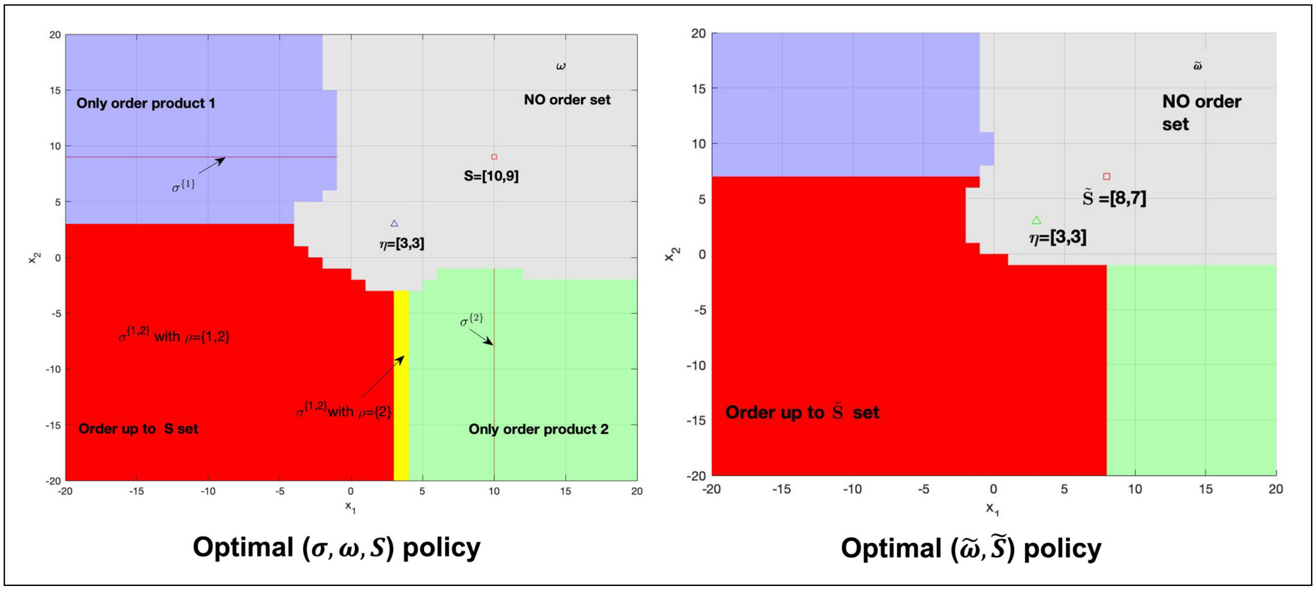

Figure 1. Illustration of the optimal policy under the base parameter set.

5.1 Illustration of the Optimal Policy

In this section, we numerically illustrate the optimal $(\sigma, \omega, \boldsymbol{S})$ policy and compare it with the $(\sigma, \boldsymbol{S})$ policy by Kalin (1980). For ease of presentation, we illustrate both policies under the infinite horizon setting with two products and the following set of parameters: $K_0 = 10, K_1 = 5, K_2 = 6, c_1 = 1.0, c_2 = 1.5, h_1 = 0.3, h_2 = 0.6, b_1 = 1.5, b_2 = 2.0,$ and $\alpha = 0.95$. The demands for the two products follow truncated normal distributions with means $d_1 = d_2 = 2$ and standard deviations $\Delta_1 = \Delta_2 = 1$ with support $[0, d_i + 3\Delta_i]$. For computational simplicity, we discretize the support and use the error tolerance $\epsilon = 0.001$. We refer to the above parameter setting as the *base parameter set*.

The left panel of Figure 1 illustrates the optimal $(\sigma, \omega, \boldsymbol{S})$ policy. Specifically, the optimal order-up-to levels are $\boldsymbol{S} = (10, 9)$ and $\boldsymbol{\eta} = (3, 3)$. The gray region is the no-order-set ω, and the blue and green regions are the order-*not*-up-to-$\boldsymbol{S}$ set, that is, $\theta \setminus \sigma$, where only product 1 is ordered for states in the blue region and only product 2 is ordered in the green region, respectively.

The order-up-to-$\boldsymbol{S}$ set σ^{ℓ} includes three subsets, corresponding to $\ell = \{1\}, \{2\}$ and $\{1, 2\}$, respectively. For $\ell = \{1\}$ and $\{2\}$, the corresponding reorder set σ^{ℓ} represents the region $\sigma^{\{1\}} = \{\boldsymbol{x}_1 | x_1 \leq -1, x_2 = S_2 = 9\}$ and the region $\sigma^{\{2\}} = \{\boldsymbol{x}_1 | x_1 = S_1 = 10, x_2 \leq -1\}$, respectively. As for $\ell = \{1, 2\}$, the reorder region $\sigma^{\{1,2\}}$ can be divided into two subsets, corresponding to the order $<_{\rho}$ where $\rho = \{2\}$ and $\{1, 2\}$, respectively. The yellow region indicates the case with $\rho = \{2\}$, and the reorder frontier set $\boldsymbol{s}^{\ell} = \{\boldsymbol{s} \mid s_1 \in (3, 4], s_2 = -3\}$. Note that for each state $\boldsymbol{x}$ in this region, we have $\boldsymbol{x} \not\leq \boldsymbol{\eta}$ (hence, $\boldsymbol{s}^{\ell} \not\leq \boldsymbol{\eta}$) and the value function $v(\boldsymbol{x})$ is increasing in $(\boldsymbol{x}, \boldsymbol{x} \vee \boldsymbol{\eta})$ along the x_2 axis (i.e., the partial order $<_{\{2\}}$), due to the $(\mathcal{K}, \boldsymbol{\eta})$-quasi-convexity of $v_t(\boldsymbol{x})$. Then, Theorem 1(4) implies that for any state in this region, it is optimal to order up to $\boldsymbol{S}$. However, for $\rho = \ell = \{1, 2\}$, the corresponding reorder frontier set is $\mathcal{F}^{\{1,2\}} = \{(-4, 3), (-3, 1), (-2, 0), (0, -1), (1, -2), (3, -3)\}$. The reorder region, highlighted in red, is

$$\sigma^{\{1,2\}} = \left\{ \boldsymbol{x} | \boldsymbol{x} \leq \boldsymbol{s} \in \mathcal{F}^{\{1,2\}} \right\}.$$

Theorem 1(4) also applies in this scenario, but according to the partial order $<_{\{1,2\}}$, that is, for states in this region, we order both products up to $\boldsymbol{S}$. The yellow and red regions together form the order-up-to $\boldsymbol{S}$ set. Hence, this example is indeed consistent with Theorem 1.

Note that Theorem 1(4) not only applies to the order-up-to-$\boldsymbol{S}$ set σ, but also applies to the order-not-up-to-$\boldsymbol{S}$ set $\theta \setminus \sigma$. In the left panel of Figure 1, the green region consists of the states under which it is optimal to order only product 2. Then, for any state $\boldsymbol{x}$ with the same x_1 in the green region, it is optimal to only order product 2 to the same inventory level, even though this inventory level is not $\boldsymbol{S}_t = (10, 9)$, based on the relationship $<_{\{2\}}$. Similarly, for states in the blue region, it is optimal to order only product 1 to the same inventory level given the same x_2. Therefore, Theorem 1(4) is also helpful in characterizing the structure of the set $\theta \setminus \sigma$.

We next compare our $(\sigma, \omega, \boldsymbol{S})$ policy with the $(\omega, \boldsymbol{S})$ policy by Kalin (1980) to highlight the impact of individual setup costs. The right panel of Figure 1 numerically shows the optimal $(\omega, \boldsymbol{S})$ policy using the base parameter set, except for zero individual setup costs, that is, $K_1 = K_2 = 0$. To avoid ambiguity, we use $\tilde{\omega}$ to denote the no-order set by Kalin (1980), and any $\boldsymbol{x} \in \{\boldsymbol{x} \mid \boldsymbol{x} \leq \tilde{\boldsymbol{S}}, \boldsymbol{x} \notin \tilde{\omega}\}$ orders up to $\tilde{\boldsymbol{S}}$, where

$\tilde{\boldsymbol{S}} = (8, 7)$ in this case. We can observe the following key differences between our $(\sigma, \omega, \boldsymbol{S})$ policy and the $(\tilde{\omega}, \tilde{\boldsymbol{S}})$ policy by Kalin (1980) from Figure 1:

(1) The optimal order-up-to-$\boldsymbol{S}$ sets are different. Without individual setup costs, Kalin (1980) shows that for any state $\boldsymbol{x}$ such that $\boldsymbol{x} \leq \tilde{\boldsymbol{S}}$, we either order up to $\tilde{\boldsymbol{S}}$, or do not order. By contrast, in the presence of individual setup costs, it may be optimal to order but not up to $\boldsymbol{S}$ for some states in the region $\{\boldsymbol{x} : \boldsymbol{x} \leq \boldsymbol{S}\}$; see the states between $\sigma^{\{1\}}$ and $\sigma^{\{1,2\}}$, and the states between $\sigma^{\{2\}}$ and $\sigma^{\{1,2\}}$ in the left panel of Figure 1. In fact, this imposes new technical challenges in showing our results.

(2) We explicitly characterize the optimal order-up-to $\boldsymbol{S}$ set, for example, σ^{ℓ} for all $\ell \subseteq \{1, 2\}, \ell \neq \emptyset$, using the partial order $<_{\rho}$ for $\rho \subseteq \ell$. Furthermore, Theorem 1 (4) suggests an ordering monotonicity property for the order-up-to-$\boldsymbol{S}$ set and the order-not-up-to $\boldsymbol{S}$ set. However, Kalin (1980) only considers the no-order-set.

(3) Due to the individual setup costs, the optimal order-up-to levels $\boldsymbol{S}$ increase, that is, $\boldsymbol{S} \geq \tilde{\boldsymbol{S}}$, and the set that orders both products to $\boldsymbol{S}$ shrinks, that is, $\sigma^{\{1,2\}} \subseteq \{\boldsymbol{x} | \boldsymbol{x} \leq \tilde{S}\} \setminus \tilde{\omega}$, while the no-order set expands, that is, $\tilde{\omega} \subseteq \omega$.

6 Bounds for the Optimal Policy

In this section, we derive the bounds for the optimal $(\sigma_t, \omega_t, \boldsymbol{S}_t)$ policy. First, we provide the upper and lower bounds for parameters of the optimal policy, for example, the bounds for the order-up-to levels $\boldsymbol{S}_t$ and for the reorder points/frontiers of the order-up-to-$\boldsymbol{S}_t$ set σ_t, and the order-not-up-to-$\boldsymbol{S}_t$ set $\theta_t \backslash \sigma_t$. Second, we construct a lower bound for the total cost by leveraging a deterministic multi-product lot-sizing system whose optimal policy is well-studied thus can be efficiently solved under the convex cost assumption. We derive these bounds for the following two main purposes. First, these bounds can be helpful in understanding the optimal $(\sigma_t, \omega_t, \boldsymbol{S}_t)$ policy. For example, we do not characterize the frontier and optimal order-up-to levels for states in the set $\theta \setminus \sigma$. Then, our bounds for the optimal parameters in Section 6.1 and Appendix F in the E-Companion provide some *rough* areas where the frontiers and order-up-to levels are located. Second, these bounds are very useful for designing and improving heuristics. For example, We utilize the optimal policy of the lower bound system as a surrogate to propose two new heuristics. We also utilize bounds for the optimal parameters to narrow down our search range when determining the order-up-to levels and reorder points in our proposed heuristics; see Section 7.1 for a detailed discussion of our heuristics.

6.1 Bounds for Optimal Parameters

In this section, we provide the upper and lower bounds for the order-up-to vector $\boldsymbol{S}_t$ and the reorder points/frontier of set σ_t, following the spirit by Veinott (1966). However, we generalize the bounds by Veinott (1966) to the multi-products and multi-setup cost setting.

Remark 8. The bounds in this section can be generalized for states in the set $\theta_t \backslash \sigma_t$, which we refer to as *index-based* bounds. As aforementioned, theses bounds can not only provide *rough area* where the order-up-to levels and reorder points/frontiers are located, but also be useful to further accelerate our proposed heuristics. However, for ease of exposition, we relegate the bounds for states in the set $\theta_t \setminus \sigma_t$ to Appendix F in the E-Companion. In all the numerical experiments, we only utilize the bounds in this section to narrow down our search regions of the optimal parameters because the demand and the inventory cost parameters are set to be stationary in each period.

We first present the following lemma that establishes the relationship between $G_t(\boldsymbol{y})$ and $g_t(\boldsymbol{y})$.

Lemma 7. *For any $t = 1, \ldots, T$, $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n$, we have the following inequalities between $G_t(\boldsymbol{y})$ and $g_t(\boldsymbol{y})$:*

(1) *For $\boldsymbol{y} \geq \boldsymbol{x}$, $G_t(\boldsymbol{y}) - G_t(\boldsymbol{x}) \geq g_t(\boldsymbol{y}) - g_t(\boldsymbol{x}) - \alpha\mathcal{K}(\boldsymbol{y} - \boldsymbol{x})$.*
(2) *For $\boldsymbol{x} \leq \boldsymbol{y} \leq \boldsymbol{x} \vee \boldsymbol{\eta}_t$, $G_t(\boldsymbol{y}) - G_t(\boldsymbol{x}) \leq g_t(\boldsymbol{y}) - g_t(\boldsymbol{x}) \leq 0$.*

Notice that, with Assumption 1, there exists $\bar{S}_{i,t}$ with $\bar{S}_{i,t} \geq \eta_{i,t}$ and $\underline{s}_{i,t}$ with $\underline{s}_{i,t} \leq \eta_{i,t}$ such that $g_{i,t}(\bar{S}_{i,t}) = g_{i,t}(\eta_{i,t}) + \alpha K_i$ and $g_{i,t}(\underline{s}_{i,t}) = g_{i,t}(\eta_{i,t}) + K_i$, for all $i \in \mathbb{N}$. Similar definitions can be found by Veinott (1966) for the single product case. Let

$$\bar{\boldsymbol{S}}_t = (\bar{S}_{1,t}, \ldots, \bar{S}_{n,t}), \tag{14}$$

$$\underline{\boldsymbol{s}}_t = (\underline{s}_{1,t}, \ldots, \underline{s}_{n,t}), \tag{15}$$

It is clear that $\underline{\boldsymbol{s}}_t \leq \boldsymbol{\eta}_t \leq \bar{\boldsymbol{S}}_t$. Now we are ready to presents our bounds.

We first derive the bounds for the order-up-to levels $\boldsymbol{S}_t$. Notice from Theorem 1 that $\boldsymbol{\eta}_t \leq \boldsymbol{S}_t$, that is, $\boldsymbol{\eta}_t$ is a natural lower bound for $\boldsymbol{S}_t$. Indeed, $\boldsymbol{\eta}_t$ can be a tight lower bound for $\boldsymbol{S}_t$, for example, $\boldsymbol{S}_T = \boldsymbol{\eta}_T$.

For an upper bound of $\boldsymbol{S}_t$, define $\mathcal{U}_t = \{\boldsymbol{U} | g_t(\boldsymbol{U}) = g_t(\boldsymbol{\eta}_t) + \alpha(K_0 + \sum_{i=1}^n K_i), \boldsymbol{U} \geq \bar{\boldsymbol{S}}_t)\}$. The following proposition shows that the boundary set $\mathcal{U}_t$ is an upper bound for $\boldsymbol{S}_t$

Proposition 1. *We have $\boldsymbol{\eta}_t \leq \boldsymbol{S}_t \leq \boldsymbol{U}$ for any $\boldsymbol{U} \in \mathcal{U}_t$.*

We next derive the lower bound and upper bounds for the order-up-to-$\boldsymbol{S}_t$ set σ_t. Recall that the order-up-to-$\boldsymbol{S}_t$ set σ_t can be characterized by a frontier $\mathcal{F}_t^{\ell}$ that consists of a set of reorder points $\boldsymbol{s}_t^{\ell}$. We shall construct two boundary sets to provide lower and upper bounds for $\mathcal{F}_t^\ell$.

$$\mathcal{LL}_t = \{\boldsymbol{U} | g_t(\boldsymbol{U}) = g_t(\boldsymbol{\eta}_t) - (K_0 + \sum_{i=1}^n K_i), \boldsymbol{U} \leq \underline{\boldsymbol{s}}_t\},$$

$$\mathcal{LU}_t = \{\boldsymbol{U} | g_t(\boldsymbol{U}) = g_t(\boldsymbol{\eta}_t) - (1-\alpha)(K_0 + \sum_{i=1}^n K_i), \boldsymbol{U} \leq \boldsymbol{\eta}_t\}.$$

The following proposition shows that the boundary sets are the lower bound and the upper bound for the frontier $\mathcal{F}_t^\ell$ and thus for σ_t, respectively.

PROPOSITION 2. *For the boundary sets $\mathcal{LL}_t, \mathcal{LU}_t$, the following properties hold,*

(1) *If $\boldsymbol{x} \leq \boldsymbol{U}$ for $\boldsymbol{U} \in \mathcal{LL}_t$, it is optimal to order up to $\boldsymbol{S}_t$.*
(2) *If $\boldsymbol{\eta}_t \geq \boldsymbol{x} \geq \boldsymbol{U}$ for $\boldsymbol{U} \in \mathcal{LU}_t$, then it is not optimal to order up to $\boldsymbol{S}_t$.*

It is worth noting that by the definition of the frontier $\mathcal{F}_T^{\mathbb{N}}$, we have $\mathcal{LL}_T \subseteq \mathcal{F}_T^{\mathbb{N}}$. That is, the boundary set $\mathcal{LL}_t$ is a tight lower bound for the reorder frontier $\mathcal{F}_T^{\mathbb{N}}$ and the order-up-to-$\boldsymbol{S}_T$ set σ_T.

6.2 Lower Bound System

In this section, to better understand the optimal policy, we provide a lower bound system for the total cost by constructing a deterministic multi-product lot-sizing counterpart. To do this, we assume that the inventory cost function $g_t(\cdot)$ is convex throughout this section, for example, under the regular linear holding and backordering costs. The convexity allows us to derive a lower bound for the expected value of $g_t(\cdot)$ by exploiting Jensen's inequality, thus leads to a lower bound system.

Note that the dynamic program in equation (6) can be equivalently formulated as follows:

$$\begin{aligned} v_1(\boldsymbol{x}_1) = & \min_{z_t \geq 0, t=1,\dots,T} \sum_{t=1}^T \alpha^{t-1} [\mathbb{E}[\mathcal{K}(z_t) + \langle \boldsymbol{c}_t, z_t \rangle \\ & + L_t(\boldsymbol{x}_t + z_t - \boldsymbol{D}_t)]] \\ \text{s.t.} \quad & \boldsymbol{x}_{t+1} = \boldsymbol{x}_t + z_t - \boldsymbol{D}_t, t = 1, \dots, T \end{aligned}$$

where we assume $\boldsymbol{c}_{T+1} = 0$ without loss. Consider the following deterministic system:

$$\begin{aligned} \underline{v}_1(\boldsymbol{x}_1) = & \min_{z_t, t=1,\dots,T} \sum_{t=1}^T \alpha^{t-1} \left[\mathcal{K}(z_t) + \langle \boldsymbol{c}_t, z_t \rangle \right. \\ & \left. + L_t \left(\boldsymbol{x}_1 + \sum_{k=1}^t z_k - \sum_{k=1}^t \boldsymbol{d}_k \right) \right], \end{aligned}$$

where we replace the random demand vector $\boldsymbol{D}_t$ in each period t with its means $\boldsymbol{d}_t$. The following proposition indicates suggests that under the convex inventory costs, we can obtain a lower bound system counterpart of the stochastic system.

PROPOSITION 3. *The deterministic system is a lower bound system for the original stochastic system, that is, $\underline{v}_1(\boldsymbol{x}_1) \leq v_1(\boldsymbol{x}_1)$ for any $\boldsymbol{x}_1$.*

Notice that the deterministic counterpart is indeed a deterministic multi-product lot-sizing problem, whose solutions and efficient algorithms have been extensively studied in the literature, for example, Zangwill (1966), Kao (1979), and Atkins and Iyogun (1988). The optimal policy for the deterministic system can be obtained via the standard lotsizing.

LEMMA 8 (Zangwill, 1966). *Under the deterministic system, if $\boldsymbol{x}_1 = 0$ then under the optimal policy we have $q_{i,t}^* x_{i,t} = 0$, and $q_{i,t}^* \in \{0, d_{i,t}, d_{i,t} + d_{i,t+1}, \dots, d_{i,t} + \cdots + d_{i,T}\}$. Moreover, the computational complexity is 2^T. If $x_{i,t} < 0$, then $q_{i,t}^* \in \{0, |x_{i,t}| + d_{i,t}, |x_{i,t}| + d_{i,t} + d_{i,t+1}, \dots, |x_{i,t}| + d_{i,t} + \cdots + d_{i,T}\}$.*

Next, we shall show that the lower bound system is asymptotically optimal as the coefficient of variations of the demands go to 0. First, we introduce the convex ordering of random variables.

DEFINITION 5. *We say that X is smaller than Y in the convex order, denoted by $X \leq_{CX} Y$ if $\mathbb{E}f(X) \leq \mathbb{E}f(Y)$ for any real convex function f given their expectations exist.*

Note that if $X \leq_{CX} Y$, we must have $\mathrm{E}[X] = \mathrm{E}[Y]$ by taking $f(x) = x$ and $f(x) = -x$, respectively. As $f(x) = x^2$ is convex, it follows that if $X \leq_{CX} Y$, then $Var(X) \leq Var(Y)$.

LEMMA 9 (Florea and Bal, 2015). *Let F_X and F_Y be the distribution functions for X and Y, respectively. Then $X \leq_{CX} Y$ if and only if $\int_x^\infty \bar{F}_X(u)du \leq \int_x^\infty \bar{F}_Y(u)du$, for all x.*

Also, notice that as the demand variation decreases to zero, the original stochastic system regresses to the deterministic counterpart. In other words, the deterministic system can be asymptotically optimal with respect to the demand variation.

PROPOSITION 4. *Let $D_{i,t}^{m+1} \leq_{CX} D_{i,t}^m$ and $\lim_{m\to\infty} D_{i,t}^m = d_{i,t}$ for all m, i, t. Let $v_1^m(\boldsymbol{x}_1)$ be the optimal cost for the stochastic system with demands $\boldsymbol{D}_t^m, t = 1, \dots, T$. Then we have $v_1^m(\boldsymbol{x}_1) \geq v_1^{m+1}(\boldsymbol{x}_1) \geq \cdots \geq \underline{v}_1(\boldsymbol{x}_1)$ and $\lim_{m\to\infty} v_1^m(\boldsymbol{x}_1) = \underline{v}_1(\boldsymbol{x}_1)$.*

It is possible to design other lower bounds for the original stochastic system using the deterministic system. Consider another deterministic system, whereby we allocate part of the joint setup cost K_0 with the value β_i to the product i such that $\sum_{i=1}^n \beta_i = 1$. We refer to the system as the independent deterministic system. It is clear that, for this system, the ordering

Table 1. Summary of all heuristics.

	Approximate θ, σ	Approximate $\boldsymbol{S}$
The independent $(\boldsymbol{s}, \boldsymbol{S})$ policy	Use rectangles determined by $\boldsymbol{s}_d$[a]	$y_i(\boldsymbol{x}) = S_{i,d}$, if $x_i < s_{i,d}$
The vector $(\boldsymbol{s}, \boldsymbol{S})$ policy	Use rectangles determined by $\boldsymbol{s}_v$	$y_i(\boldsymbol{x}) = S_{i,v}$, if $\boldsymbol{x} <_{\mathbb{N}} \boldsymbol{s}_v$
The linear interpolation $(\boldsymbol{s}, \boldsymbol{S})$ policy	Use lines/hyperplanes to approximate frontiers of σ^{ℓ}	$y_i(\boldsymbol{x}) = S_{i,l}$, if $\boldsymbol{x} \in \Sigma^{\ell}, i \in \ell$, where frontiers of Σ^{ℓ} are linear
The deterministic approximation policy	Use the ordering sets of the deterministic system	$y_i(\boldsymbol{x}) = S_{i,A}$, if $\hat{y}_i(\boldsymbol{x}) > x_i$[b]
The weighted deterministic approximation policy	Use the ordering sets of the deterministic system	$y_i(\boldsymbol{x}) = S_{i,W}$, if $\hat{y}_i(\boldsymbol{x}) > x_i$, where $S_{i,W} = \beta^*\hat{y}_i(\boldsymbol{x}) + (1-\beta^*)\bar{S}_i$[c].

[a] $\boldsymbol{s}_d, \boldsymbol{s}_v, \boldsymbol{S}_d, \boldsymbol{S}_v, \boldsymbol{S}_l, \boldsymbol{S}_A$ are determined by a greedy algorithm in Appendix B in the E-Companion.
[b] $\hat{\boldsymbol{y}}(\boldsymbol{x})$ is the optimal order up to level of the deterministic system given $\boldsymbol{x}$.
[c] $\bar{\boldsymbol{S}}$ is an upper bound of $\boldsymbol{S}$ defined in equation (14), and $\beta^* \in (0, 1)$ is determined by an exhausted search method.

decisions for different products are independent. Then

$$\hat{v}_1(\boldsymbol{x}_1, \boldsymbol{\beta}) \equiv \sum_{i=1}^{n} \min_{z_{i,t}, t=1,\ldots,T} \sum_{t=1}^{T} \alpha^{t-1} \left[\delta(z_{i,t})(\beta_i K_0 + K_i) + c_i z_{i,t} + L_{i,t}\left(x_{i,1} + \sum_{k=1}^{t} z_{i,k} - \sum_{k=1}^{t} d_{i,k} \right) \right].$$

The above optimization program is simple as we can decide the optimal policy for each product independently. Thus, we only need to solve n one-dimensional dynamic programs for the independent deterministic system. As the independent deterministic system incurs less fixed costs for the deterministic system under the same ordering policy, we thus have the following result.

Proposition 5. *We have* $\underline{v}_1(\boldsymbol{x}_1) \geq \hat{v}_1(\boldsymbol{x}_1, \boldsymbol{\beta})$ *and as a result* $v_1(\boldsymbol{x}_1) \geq \max_{\boldsymbol{\beta} \in \Delta} \hat{v}_1(\boldsymbol{x}_1, \boldsymbol{\beta})$, *where* $\Delta = \{\boldsymbol{\beta} : \boldsymbol{\beta} \geq 0, \sum_{i=1}^{n} \beta_i = 1\}$.

7 Heuristics and Numerical Studies

Although we show the optimality of the $(\sigma, \omega, \boldsymbol{S})$ policy, the optimal policy is difficult to implement in practice because of its complex nature, especially when we have a large number of products. To tackle this issue, we propose five simple heuristics to approximate the optimal $(\sigma, \omega, \boldsymbol{S})$ policy and test their performance. These heuristics are in part motivated by the structural properties and bounds of the optimal $(\sigma, \omega, \boldsymbol{S})$ policy analyzed in the previous sections, and all our heuristics turn out to be effective.

7.1 Simple and Effective Heuristics

In this section, we propose five simple heuristics to approximate the optimal $(\sigma, \omega, \boldsymbol{S})$ policy. Recall that σ is the order-up-to-$\boldsymbol{S}$ set, θ is the ordering set including σ, and $\boldsymbol{S}$ is the optimal order-up-to level. Inspired by the structural properties and bounds of the $(\sigma, \omega, \boldsymbol{S})$ policy, each heuristic approximates these parameters in different ways.

When approximating θ or σ, the first two heuristics use hyper-rectangles, while the third heuristic uses hyper-planes to approximate the frontier of σ. The rest two heuristics, however, utilize the lower bound system to determine when to order. That is, we first fully solve the lower bound system, which is a multi-product lot-sizing problem, then follow the re-order behavior of the lower bound system. To approximate $\boldsymbol{S}$, we use a local greedy algorithm to determine $\boldsymbol{S}$ in the first four heuristics (see Table EC.1 in Appendix B). By contrast, we utilize the order-up-to levels of the lower bound system to construct a weighted average to approximate $\boldsymbol{S}$ in the fifth heuristic. When designing the hyper-rectangles, hyper-planes, the search regions of the greedy algorithm, and the order-up-to levels, we make use of the bounds proposed in the previous sections. See Table 1 for a summary and Figure 2 for a graphical illustration of all heuristics with two products.

We next illustrate the heuristics in detail.

- **The independent $(\boldsymbol{s}, \boldsymbol{S})$ policy**. This policy ignores the impacts of the joint setup cost K_0 and replenishes each product independently. Specifically, in each period, there exist two constant vectors $\boldsymbol{s}_d = (s_{1,d}, \ldots, s_{n,d})$ (the reorder points) and $\boldsymbol{S}_d = (S_{1,d}, \ldots, S_{n,d})$ (the order-up-to levels) with $s_{i,d} < S_{i,d}$ for each product i. Then, if the net inventory level x_i of product i is smaller than $s_{i,d}$, order for product i to raise its inventory level up to $S_{i,d}$; otherwise, do not order. Hence, the order quantity q_i for product i is $q_i = \begin{cases} S_{i,d} - x_i & \text{if } x_i < s_{i,d}, \\ 0 & \text{otherwise.} \end{cases}$

  One can see from Figure 2 that this policy approximates σ by hyper-rectangles determined by the reorder points $\boldsymbol{s}_d$. Also, note that this policy captures the ordering monotonicity property, that is, ordering product i up to $S_{i,d}$ for all $x_i < s_{i,d}$ given the states of other products. It is intuitive that such a policy can be effective when the joint setup cost is small.

  Clearly, choosing parameters $\boldsymbol{s}_d$ and $\boldsymbol{S}_d$ is crucial for the independent $(\boldsymbol{s}, \boldsymbol{S})$ policy. In numerical experiments, we design a local grid algorithm to search

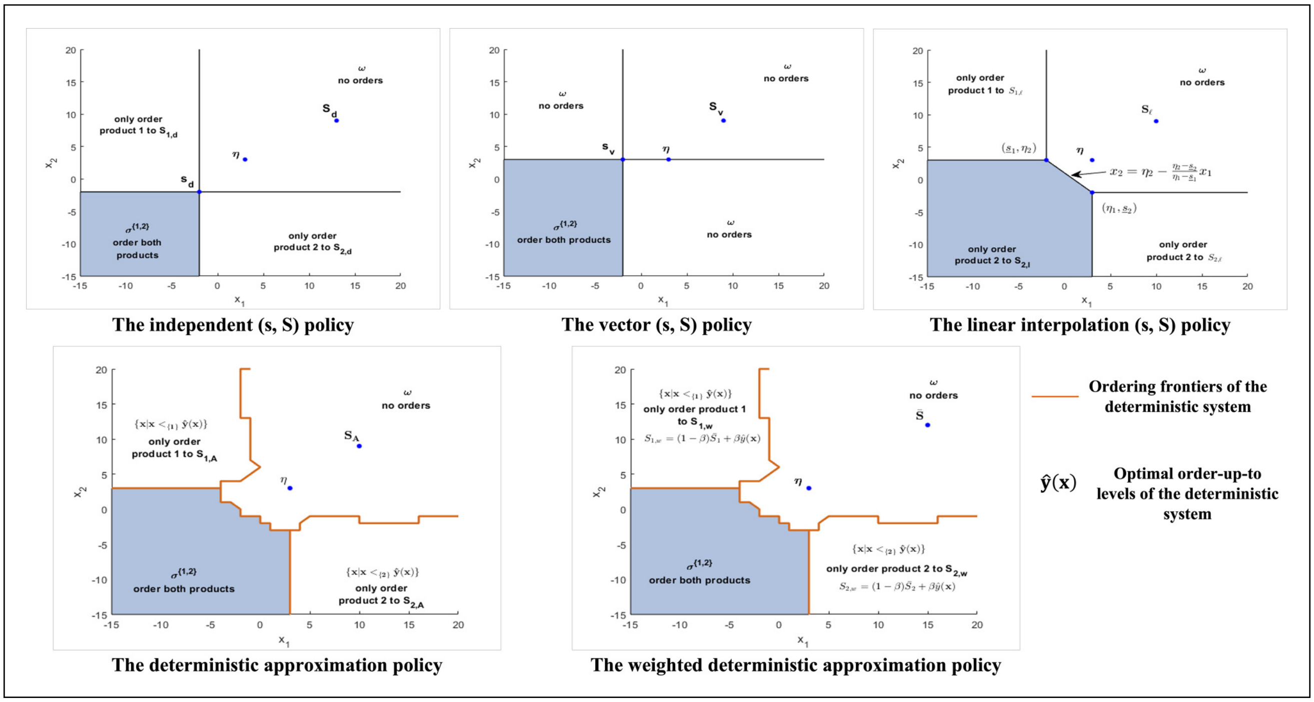

Figure 2. Graphical illustration of heuristics with two products using the base parameter set.

for the optimal parameters and utilize the bounds in the previous sections to narrow the search region and speed up the search (see Table EC.1 in Appendix B[1]). Also note that we adopt the greedy algorithm for the ease of implementation, and one can improve the performance by designing better algorithms to determine the parameters.

- **The vector (s, S) policy**. This policy only accommodates joint ordering: whenever we order, we order all products simultaneously. Specifically, in each period, there exists two vectors $\boldsymbol{s}_v$ (the reorder point) and $\boldsymbol{S}_v$ (the order-up-to levels) with $\boldsymbol{s}_v < \boldsymbol{S}_v$. If the inventory state $\boldsymbol{x}$ is component-wise smaller than $\boldsymbol{s}_v$, then order for all the products and raise the inventory levels up to $\boldsymbol{S}_v$; otherwise, order nothing. Hence the order quantity q_i for product i is $q_i = \begin{cases} S_{i,v} - x_i & \text{if } \boldsymbol{x} < \boldsymbol{s}_v, \\ 0 & \text{otherwise.} \end{cases}$

  Essentially, the vector $(\boldsymbol{s}, \boldsymbol{S})$ policy approximates the reorder region σ by $\{\boldsymbol{x}|\boldsymbol{x} < \boldsymbol{s}_v\}$ and ignores the ordering outside σ. We also use the greedy algorithm to find the parameters $\boldsymbol{s}_v$ and $\boldsymbol{S}_v$.
- **The linear interpolation (s, S) policy**. This policy is inspired by the structure of the optimal $(\sigma, \omega, \boldsymbol{S})$ policy. Note that to approximate σ, it is sufficient to approximate its frontier, that is, $\mathcal{F}_t^\ell$. To do this, we use hyper-planes to approximate the frontier. For ease of implementation, we use $\boldsymbol{\eta}$ and the reorder points in equation (15) to construct the linear/affine frontier, and use the greedy algorithm to search for the order-up-to levels $\boldsymbol{S}$.

  Specifically, this policy can be described as follows: in each period, there are two vectors $\underline{\boldsymbol{s}}$ defined in equation (15) in Section 6 and $\boldsymbol{S}_l$ with $\boldsymbol{\eta} \le \boldsymbol{S}_l$. Define $\Sigma^\ell = \{(x_1, \ldots, x_n)|x_i < \eta_i, i \in \ell, x_j \ge \eta_j, j \in \mathbb{N} \setminus \ell, \sum_{i=1}^{|\ell|} a_i x_i \le \sum_{i=1}^{|\ell|} a_i \underline{s}_i\}$, where $\ell \subseteq \mathbb{N}$ and $(a_1, \ldots, a_{|\ell|})$ are constants among which at least one element is nonzero. Then, $q_i = \begin{cases} S_{i,l} - x_i & \text{if } \boldsymbol{x} \in \Sigma^\ell, i \in \ell, \\ 0 & \text{otherwise.} \end{cases}$

  That is, we use hyper-planes $\sum_{i=1}^{|\ell|} a_i x_i \le \sum_{i=1}^{|\ell|} a_i \underline{s}_i$ to approximate the frontier of the reorder region σ^ℓ in the optimal $(\sigma, \omega, \boldsymbol{S})$ policy; see Figure 2 for an illustration of two products. We also present an algorithm to determine the reorder point $\underline{\boldsymbol{s}}$ of the linear interpolation $(\boldsymbol{s}, \boldsymbol{S})$ policy based on our analysis in Section 6; see Table EC.2 in Appendix B. Note that the linear interpolation $(\boldsymbol{s}, \boldsymbol{S})$ policy resembles the ordering frontier of the one-period deterministic problem with two products, which is indeed a linear segment (see Appendix C for a detailed discussion in the E-Companion).
- **The deterministic approximation policy**. This heuristic utilizes the deterministic counterpart of the stochastic system, that is, the lower bound system in Section 6.2, to determine whether to order for each inventory state. For instance, let $\hat{\boldsymbol{y}}(\boldsymbol{x})$ denote the optimal order-up-to levels for state $\boldsymbol{x}$ under the

deterministic system. If it is optimal to order for the product i in the set ℓ in the deterministic system for state $\boldsymbol{x}$, that is, $\hat{y}_i > x_i$, then we also order for product i and bring its inventory level to $S_{i,A}$ in the stochastic system, where $\boldsymbol{S}_A$ is determined by a local grid search algorithm (see Table EC.1 in Appendix B). The idea here is to utilize the ordering behavior of the lower system as a surrogate for the stochastic system, because of the asymptotic optimality of the lower bound system. Indeed, in the two-product example in Appendix D in the E-Companion, we numerically solve the lower bound system and find that its ordering frontiers are very close to the counterparts of the optimal $(\sigma, \omega, \boldsymbol{S})$ policy. Specifically, this policy can be described as follows: in each period,

Step 1. Solve the deterministic system and record the corresponding optimal order-up-to levels $\hat{\boldsymbol{y}}(\boldsymbol{x})$ for each inventory state $\boldsymbol{x}$;

Step 2. Initialize $\boldsymbol{S}_A = \boldsymbol{S}^0$;

Step 3. Compute the value function $V(\boldsymbol{S}_A)$ according to the following ordering policy: for the inventory state $\boldsymbol{x}$, order $q_i^0(x_i) = \begin{cases} S_{i,A} - x_i & \text{if } \hat{y}_i > x_i \\ 0 & \text{otherwise} \end{cases}$ for product i;

Step 4. Find another $\boldsymbol{S}' \neq \boldsymbol{S}^0$ in the local region of $\boldsymbol{S}^0$ and compute the corresponding value function $V(\boldsymbol{S}')$ using the same ordering policy in step 2;
if the mean of $V(\boldsymbol{S}') - V(\boldsymbol{S}_A)$ is negative $\boldsymbol{S}_A = \boldsymbol{S}', V(\boldsymbol{S}_A) = V(\boldsymbol{S}')$; continue to search $\boldsymbol{S}'$;else break;Return $\boldsymbol{S}_A = \boldsymbol{S}', V_A = V(\boldsymbol{S}_A)$ and $\boldsymbol{y}_A^*(\boldsymbol{x}) = \begin{cases} S_{i,A} & \text{if } \hat{y}_i(\boldsymbol{x}) > x_i, \\ 0 & \text{otherwise.} \end{cases}$

One advantage of this heuristic is that it can approximate not only σ, but also $\theta \setminus \sigma$ and ω. This is because we can fully solve the deterministic counterpart which is a multi-product lot sizing problem. Although this heuristic has relatively larger (but tolerable) computation burden compared with the previous heuristics, one can expect better performance. Indeed, this heuristic performs extremely well in our numerical studies, whose average performance gap is $< 1\%$.

- **The weighted deterministic approximation policy**. Similar to the previous policy, this heuristic also utilizes the deterministic counterpart to approximate the ordering regions. By sharp contrast, this heuristic approximates the order-up-to levels by constructing a weighted average using the optimal order-up-to levels of the deterministic system and the upper bound of $\boldsymbol{S}$. For instance, let $\hat{\boldsymbol{y}}(\boldsymbol{x})$ denote the optimal order-up-to levels for state $\boldsymbol{x}$ under the deterministic system. If it is optimal to order for the product i in the set ℓ in the deterministic system, that is, $\hat{y}_i > x_i$, then we order for product i and bring its inventory level to $S_{i,W}(\boldsymbol{x}) = \beta\hat{y}_i + (1-\beta)\bar{S}_i$ in the stochastic system, where $\beta \in (0,1)$ and $\bar{\boldsymbol{S}}$ is an upper bound of the optimal order-up-to levels $\boldsymbol{S}$ defined in equation (14). Clearly, $\boldsymbol{S}_W >_\ell \boldsymbol{x}$. Specifically, this policy can be described as follows: in each period,

Step 1. Solve the deterministic system and record the corresponding optimal order-up-to levels $\hat{\boldsymbol{y}}(\boldsymbol{x})$ for each inventory state $\boldsymbol{x}$;

Step 2. For $\beta = 0.05, \ldots, 1$,calculate the order-up-to levels $\boldsymbol{y}(\boldsymbol{x}, \beta) = \begin{cases} \beta\hat{y}_i(\boldsymbol{x}) + (1-\beta)\bar{S}_i & \text{if } \hat{y}_i(\boldsymbol{x}) > x_i \\ x_i & \text{otherwise} \end{cases}$,compute the value function $V^*(\beta)$ for each β;

Step 3. Return $\beta^* = \arg\min_\beta V^*(\beta)$, $V_W = V^*(\beta^*)$ and $\boldsymbol{S}_w(\boldsymbol{x}) = \boldsymbol{y}(\boldsymbol{x}, \beta^*)$.

The idea to create a weight average approximate of order-up-to levels is in part inspired by our numerical experiments of the deterministic system. In the numerical example in Appendix D in the E-Companion, we further observe that the optimal order-up-to levels of the deterministic system for most inventory states can serve as a lower bound of the stochastic one. Therefore, we approximate the optimal order-up-to levels by a linear combination of their deterministic counterpart and some upper bounds. The heuristic also exhibits excellent performance in our numerical experiments.

7.2 Numerical Results

In this section, we numerically test the performance of each heuristic proposed in Section 7.1, using the same infinite-time horizon setting with two products as we illustrate the optimal $(\sigma, \omega, \boldsymbol{S})$ policy in Section 5.1. To evaluate the performance of each heuristic, we measure the percentage of the performance gap against the optimal $(\sigma, \omega, \boldsymbol{S})$ policy, that is,

$$\xi_p = \frac{V_p - V^*}{V^*} \times 100\%, p \in \{d, v, l, A, W\},$$

where V^* denotes the value function of the optimal $(\sigma, \omega, \boldsymbol{S})$ policy, the subscripts $\{d, v, l, A, W\}$ correspond to the independent $(\boldsymbol{s}, \boldsymbol{S})$ policy, the vector $(\boldsymbol{s}, \boldsymbol{S})$ policy, the linear interpolation $(\boldsymbol{s}, \boldsymbol{S})$ policy, the deterministic approximation policy, and the weighted deterministic approximation policy, respectively, and V_p denotes the value function of the corresponding heuristic.

To isolate the impact of each parameter, we evaluate the performance of each heuristics by varying only one parameter in the base parameter set. Specifically, we set parameters $K_0 \in \{6, 8, 10, 12, 14, 20\}, K_1 = \{3, 4, 6, 7, 8\}, K_2 \in \{4, 5, 8, 9, 10\}, b_1 \in \{1.4, 1.6, 1.8, 2, 2.5\}, h_2 \in \{0.4, 0.5, 0.7,$

$0.8, 1\}, c_2 \in \{1.2, 1.3, 1.4, 1.6, 1.8\}, d_i \in \{1, 3, 4, 5, 6\}, \Delta_i \in \{0.5, 1.5, 2, 2.5, 3\}$.

We report the average performance gap across all states of each heuristic against the optimal value function, denoted by $\bar{\xi}_p$, $p \in \{d, v, l, A, W\}$. We also report the performance gaps for five particular states, that is, $\boldsymbol{x}_0 = (-5, -5), (-1, 10), (10, -1), (5, 5)$, and the origin (0,0). We purposefully choose these states because $(-5, -5)$ belongs to σ, (−1,10) and (10,−1) belong to $\theta \backslash \sigma$, for which we only order for product 1 or product 2, respectively, and $(5, 5)$ belongs to ω, in all numerical experiments. This allows us to investigate the performance of each heuristic in different ordering regions. We conduct all numerical experiments using Matlab version 2019a on HP Z2 Tower G4 workstation with CPU i7-9700 (3.0 GHz) and RAM 327768 MB. For ease of presentation, we relegate all numerical results to Appendix E; see Tables EC.3 to EC.5 for the impacts of all parameters on the average performance gap of each heuristic and Tables EC.6 and EC.7 for results on the performance gap of each heuristic with respect to states $\boldsymbol{x}_0 = (-5, -5), (-1, 10), (10, -1), (5, 5)$ and the origin (0,0).

Below, we summarize some interesting observations from the numerical results.

- The last three heuristics perform extremely well and in particular the weighted deterministic approximation policy dominates in almost all cases. From Tables EC.3 to EC.5, we have the following ordering among five heuristics, that is, $\bar{\xi}_W < \bar{\xi}_A < \bar{\xi}_l < \bar{\xi}_d < \bar{\xi}_v$, in almost all of our numerical experiments. In particular, the average performance gaps of the deterministic approximation policy and the weighted deterministic approximation policy are both < 1% ($\bar{\xi}_W < 0.726\%, \bar{\xi}_A < 0.771\%$), whereas the average performance gap of the linear interpolation $(\boldsymbol{s}, \boldsymbol{S})$ policy is < 3.58% ($\bar{\xi}_l < 3.58\%$). However, the average performance gaps of the independent $(\boldsymbol{s}, \boldsymbol{S})$ policy and the vector $(\boldsymbol{s}, \boldsymbol{S})$ policy are > 5% and 10% in most cases, respectively. The reason why the last three heuristics perform well is because they provide close approximations on the ordering sets. Note that the optimal ordering frontier of the stochastic system and its deterministic counterpart are quite close with each other (see the example in Figure EC.2 in Appendix D). This implies that the optimal ordering regions of both system including θ, σ and ω only differ in a few inventory states, leading to an excellent performance of the (weighted) deterministic approximation policy. The linear interpolation $(\boldsymbol{s}, \boldsymbol{S})$ policy also provides good approximation using lines or hyperplanes. It is also worth noting that although dominated by the (weighted) deterministic approximation policy, the linear interpolation $(\boldsymbol{s}, \boldsymbol{S})$ policy is much easier to implement because it does not need to fully solve the deterministic system.
- The last three heuristics are much more robust than the independent/vector $(\boldsymbol{s}, \boldsymbol{S})$ policy. In terms of average performance, the performance gap of the linear interpolation $(\boldsymbol{s}, \boldsymbol{S})$ policy varies from 0.49% to 3.57%, that is, $\bar{\xi}_\ell \in [0.49\%, 3.57\%]$, whereas that of the (weighted) deterministic approximation policy is < 1%. By contrast, the independent vector $(\boldsymbol{s}, \boldsymbol{S})$ policy has a performance gap between 5.18% and 11.96%, that is, $\bar{\xi}_d \in [5.18\%, 11.96\%]$, while the vector $(\boldsymbol{s}, \boldsymbol{S})$ policy has a gap between 2.21% and 26.11%, that is, $\bar{\xi}_v \in [2.21\%, 26.11\%]$. In terms of different states (or ordering regions), all heuristics have relatively stable performances, except for the vector $(\boldsymbol{s}, \boldsymbol{S})$ policy. Note that the vector $(\boldsymbol{s}, \boldsymbol{S})$ policy performs quite well for the order-up-to-$\boldsymbol{S}$ set σ and no order set ω, for example, (ξ_v=0.25% for $\boldsymbol{x}_0 =$ (-5,-5) and $\boldsymbol{x}_0 = (5, 5)$ when $d_i = 6$), but performs poorly for the order-not-up-to-$\boldsymbol{S}$ set $\theta \backslash \sigma$, for example, (ξ_v=18.41% for $\boldsymbol{x}_0$=(-1,10) and ξ_v=36.93% for $\boldsymbol{x}_0 = (5, 5)$ when $d_i = 1$). It is intuitive since the vector $(\boldsymbol{s}, \boldsymbol{S})$ policy only orders all products simultaneously and ignores individual ordering for each product.
- The joint setup cost K_0 and the mean demand d_i have different impacts on the performances of all heuristics. The performances of the linear interpolation $(\boldsymbol{s}, \boldsymbol{S})$ policy and the independent $(\boldsymbol{s}, \boldsymbol{S})$ policy improve as the joint setup cost and mean demand decrease. By contrast, the increases of the joint setup cost and the mean demand can improve the performance of the vector $(\boldsymbol{s}, \boldsymbol{S})$ policy and the (weighted) deterministic approximation policy. As for other parameters, the increase of the variable cost c_2 can mildly improve the performance of all heuristics while other parameters do not exhibit any monotonic impacts.

In summary, we recommend the last three heuristics, that is, the linear interpolation $(\boldsymbol{s}, \boldsymbol{S})$ policy, the deterministic approximation policy, and the weighted deterministic approximation policy, to manage complex inventory systems with multiple setup costs because their simplicity and efficiency. Specifically, if the problem's deterministic counterpart is not difficult to solve, then the (weighted) deterministic approximation policy is strongly recommended, otherwise, the linear interpolation $(\boldsymbol{s}, \boldsymbol{S})$ policy may be the best choice.

8 Extensions

In this section, we briefly discuss how we can extend our results to systems with time-varying ordering costs, more complex setup costs and fixed lead-times.

8.1 Time-Varying Setup Costs

Consider the system with the same setting as the counterpart in Section 2 except for having time-varying ordering costs. Denote by $c_t(\boldsymbol{z})$ be the ordering costs in period t for $\boldsymbol{z} = (z_1, \ldots, z_n)$, the vector of order quantities. Then, $c_t(\boldsymbol{z})$ is defined as follows:

$$c_t(\boldsymbol{z}) = K_{0t}\delta\left(\sum_{i=1}^{n} z_i\right) + \sum_{i=1}^{n}[K_{i,t}\delta(z_i) + c_{i,t}z_i].$$

We define $\mathcal{K}_t(z) = K_{0t}\delta\left(\sum_{i=1}^n z_i\right) + \sum_{i=1}^n K_{i,t}\delta(z_i)$. If $\mathcal{K}_t \leq \alpha\mathcal{K}_{t+1}$, then by property (2) of Lemma 2, we can still show that v_t is $(\mathcal{K}_t, \boldsymbol{\eta})$-quasi-convex. Hence, the structure of the optimal policies for such system can still be described as $(\sigma_t, \omega_t, \boldsymbol{S}_t)$ policies.

8.2 General Setup Costs

We can also extend our result to more complicated multiple setup costs. Consider the case where there is a (or a series of) partial joint setup cost(s) for a set of products $K_\ell\delta(\sum_{i\in\ell} z_i)$ where $K_\ell \geq 0, \ell \in 2^{\mathbb{N}}$, in addition to the joint setup cost K_0 and individual setup cost $K_i, i \in \mathbb{N}$. For example, perhaps the products in the set ℓ require special containers during transportation.

Note that in the inductive proof of Lemma 6, we only utilize the subadditivity (see Lemma 1) and the nondecreasing property of $\mathcal{K}(\cdot)$. However, it is clear that all setup costs are nondecreasing. Hence our results hold for any subadditive function $\mathcal{K}$. It follows that with $K_\ell\delta(\sum_{i\in\ell} z_i)$ for some $\ell \in 2^{\mathbb{N}}$, the generalized indicator function $\mathcal{K}(\cdot)$ is still subadditive, which implies that, the optimal inventory policy is a $(\sigma, \omega, \boldsymbol{S})$ policy in this case.

Furthermore, we can show the $(\mathcal{K}, \boldsymbol{\eta})$-quasi-convexity of value functions of systems with quantity-dependent setup cost, that is,

$$\mathcal{K}_t(z) = K_0\delta\left(\sum_{i=1}^n z_i\right) + \sum_{i=1}^n \kappa_i(z_i),$$

where $\kappa_i(z_i) = \begin{cases} 0 & \text{if } z_i = 0, \\ K_i\delta(z_i) + mK_i & \text{if } \lceil \frac{z_i}{d_i} \rceil = m, \text{ for } z_i > 0. \end{cases}$

Hence, $d_i > 0$ can be viewed as the batch size for product i and $\lceil x \rceil$ denotes the smallest integer larger than or equal to $\boldsymbol{x}$. This type of cost is common in real-world logistics and transportation services. It is clear that the above generalized indicator function $\mathcal{K}(\cdot)$ is subadditive and nondecreasing. As a consequence, the $(\mathcal{K}, \boldsymbol{\eta})$-quasi-convexity of the value function is preserved. However, it is challenging to characterize the structure of optimal policies in this setting, and the $(\sigma, \omega, \boldsymbol{S})$ policies may not be optimal. The reason is that the setup costs for individual products are now functions of the ordering quantities. Nevertheless, the $(\mathcal{K}, \boldsymbol{\eta})$-quasi-convexity of value functions may still be useful to analyze the structure of the optimal policy and design heuristics.

9 Concluding Remarks

In this article, we investigate the optimal procurement strategy for a finite-horizon, periodic-review inventory system with multiple products and multiple setup costs. The setup costs consist of a joint setup cost, an individual setup cost for each product, and a setup cost for ordering a subset of products. The objective is to minimize the total discounted expected cost.

Under our assumptions, which cover situations when demands are stochastically increasing over time, we show that a $(\sigma, \omega, \boldsymbol{S})$ policy is optimal for the above system: we order up to $\boldsymbol{S}$ for any inventory state in the set σ, but do not order for states in the set ω, and we order up to certainty quantities (but not up to $\boldsymbol{S}$) for states neither in σ nor in ω. To prove the results, we propose a class of $(\mathcal{K}, \boldsymbol{\eta})$-quasi-convex functions that generalizes Veinott (1966) and Kalin (1980). We provide the ordering monotonicity properties for the optimal $(\sigma, \omega, \boldsymbol{S})$ policy by exploiting partial orders, which is absent from the prior literature. We also provide bounds for the optimal policy, which are new for our setting. Under the convex inventory costs, we derive a lower bound system which is asymptotically optimal as the coefficient of variation of demands decreases to zero.

These structural properties and bounds inspire us to design five simple and effective heuristics that are easy to be implemented in practice: the independent $(\boldsymbol{s}, \boldsymbol{S})$ policy, the vector $(\boldsymbol{s}, \boldsymbol{S})$ policy, the linear interpolation $(\boldsymbol{s}, \boldsymbol{S})$ policy, the deterministic approximation policy, and the weighted deterministic approximation policy. The first two heuristics use hyper-rectangles to approximate the ordering sets whereas the linear interpolation $(\boldsymbol{s}, \boldsymbol{S})$ policy uses lines/hyperplanes to approximate the frontiers of all ordering sets. In contrast, the deterministic approximation policy and the weighted deterministic approximation policy first solve an lower bound system, then utilize its ordering sets as a surrogate as that of the optimal $(\sigma, \omega, \boldsymbol{S})$ policy. We also design greedy and exhausted search algorithms to specify the parameters of each heuristics.

We then numerically illustrate the optimal $(\sigma, \omega, \boldsymbol{S})$ policy and investigate the performance of all heuristics. Extensive numerical experiments suggest that the last three heuristics perform extremely well and the weighted deterministic approximation policy dominates others in almost all numerical experiments. The average performance gap of the linear interpolation $(\boldsymbol{s}, \boldsymbol{S})$ policy is $< 3.58\%$, whereas the average performance gaps of the deterministic approximation policy and the weighted deterministic approximation policy are both $< 1\%$ in all experiments. Hence, we recommend these three heuristics to manage complex inventory systems with multiple setup costs because their simplicity and efficiency. Specifically, if the problem's deterministic counterpart is not difficult to solve, then the (weighted) deterministic approximation policy is strongly recommended, otherwise, the linear interpolation $(\boldsymbol{s}, \boldsymbol{S})$ policy may be the best choice.

Finally, we discuss how to generalize our results to time-varying and general subadditive setup costs.

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Note

1. We adopt this algorithm due to the computational burden. Note that the state space in our numerical experiments is no less than 30-by-30, for example, $[-15:15]\times[-15:15]$, hence, exhaustive search requires more than 810,000 times dynamic programming for each loop.
2. Science

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